

Network Design for ALPHA LOGISTICS

USMAN AKHTER
24799861

Contents

ABSTRACT	3
INTRODUCTION	3
AIM AND OBJECTIVES	3
NETWORK DESIGN	4
NETWORK OVERVIEW	4
HARDWARE AND TOPOLOGY	8
IP ADDRESSING AND SUBNETTING (CIDR)	9
VLAN SEGMENTATION DESIGN	10
INTER-VLAN ROUTING AND DYNAMIC ROUTING - EIGRP	11
DHCP DESIGN	13
DNS AND WEB SERVICE DESIGN	14
EMAIL SERVICE DESIGN	15
WIRELESS NETWORK DESIGN (STAFF WI-FI VLAN)	16
SECURITY DESIGN	18
NETWORK CONFIGURATION TESTS	20
INTER-VLAN COMMUNICATION TEST	20
INTER-SITE CONNECTIVITY TEST	23
DHCP SERVER TEST	25
WIRELESS CONNECTIVITY TEST	27
WEB AND DNS SERVER TEST	29

EMAIL SERVICE TEST	32
ROUTER CONFIGURATION TEST	34
NETWORK EVALUATION AND DISCUSSION	36
CONCLUSION	37
APPENDIX A	37
REFERENCES	38

Abstract

As a network Engineer, I have designed a network that meets the needs of Alpha logistics who operate 4 offices and a factory across Manchester and London. My aim is to design a network that is secure, scalable and reliable network that has end-to-end communication, DHCP, EIGRP, CIDR, application layer services and VLAN segmentation. This report demonstrates the proposed design and provides an efficient and cost-effective solution through tested to ensure its functionality

Introduction

I have designed a medium level network for logistics business called ALPHA-Logistics. The client has 50 employees in various department like sales, marketing, HR, IT and production management across Manchester and London offices each city has two offices and one factory in Manchester.

As the company continues to expand its operations, its existing network infrastructure is no longer compatible to support growing demands of a logistics business. As a network engineer, I'd suggest that they need a new scalable and reliable network design to improve performance, enhance security and ensure high availability of critical business operations like email and web service

The network has been designed using Cisco Packet Tracer, which used enterprise networking technologies such as VLAN segmentation, inter-VLAN routing, dynamic routing (EIGRP), DHCP, DNS, wireless networking and network security through ssh, devices hardening and access control. I have implemented full IPv4 addressing, centralised server services and hub-and-spoke WAN topology with the City of London office acting as main branch/HQ.

The report will outline a complete network solution, covering the design methodology, IP addressing scheme, routing algorithm, switch configuration, and security implementation. The conclusion will evaluate overall effectiveness of the design against the client requirements and will provide recommendations for future growth.

Aim and Objectives

My primary objective is to design and implement a secure, scalable and reliable enterprise network infrastructure for ALPHA-LOGISTICS that supports communication between multiple offices and a factory while meeting business expectations and technical requirements

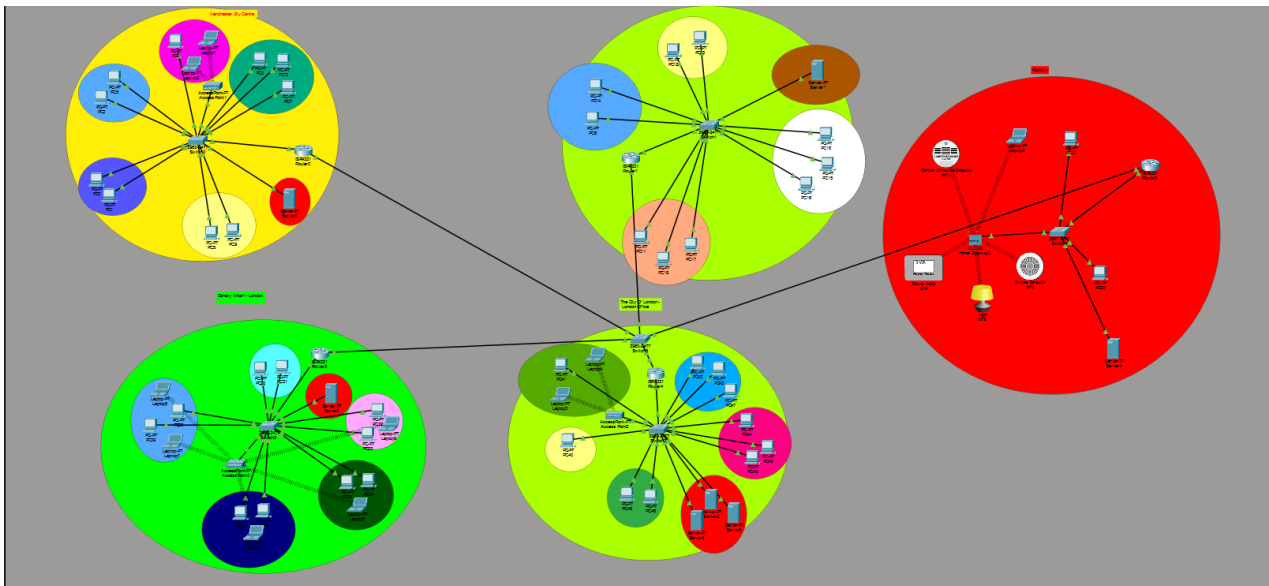
Objectives:

- Develop an IP addressing scheme using CIDR and subnetting.
- Design VLANs to logically separate departments and improve security.
- Configure dynamic routing between all sites using EIGRP.
- Implement core network services including DHCP, DNS, HTTP, and Email.
- Deploy secure wireless access for staff.
- Apply network security measures such as SSH and device hardening.
- Test and validate end-to-end connectivity and performance

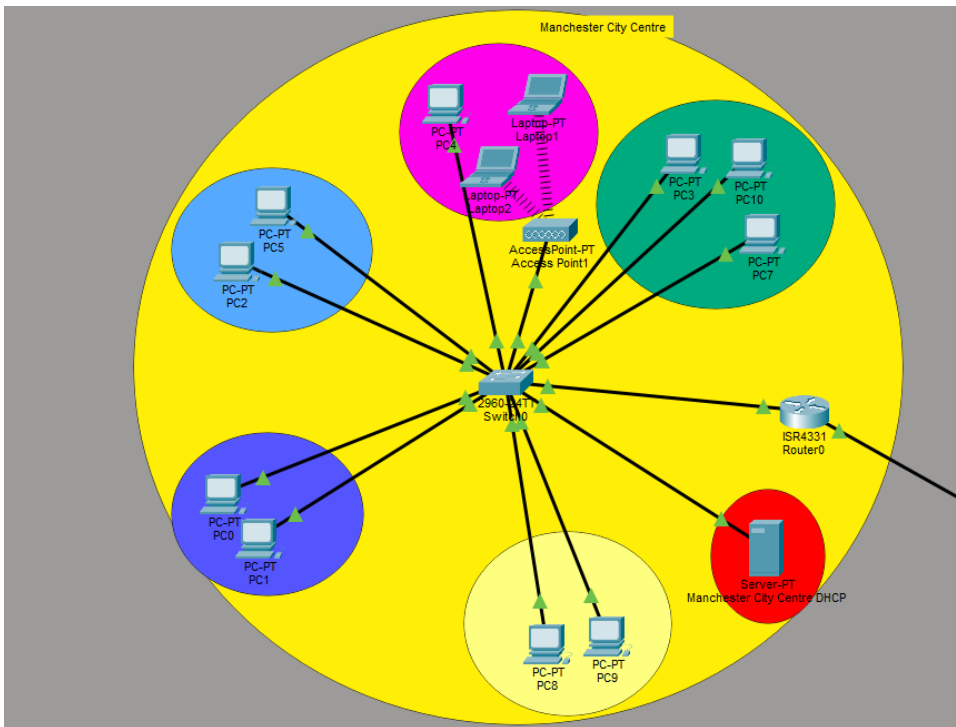
Network Design

Network overview

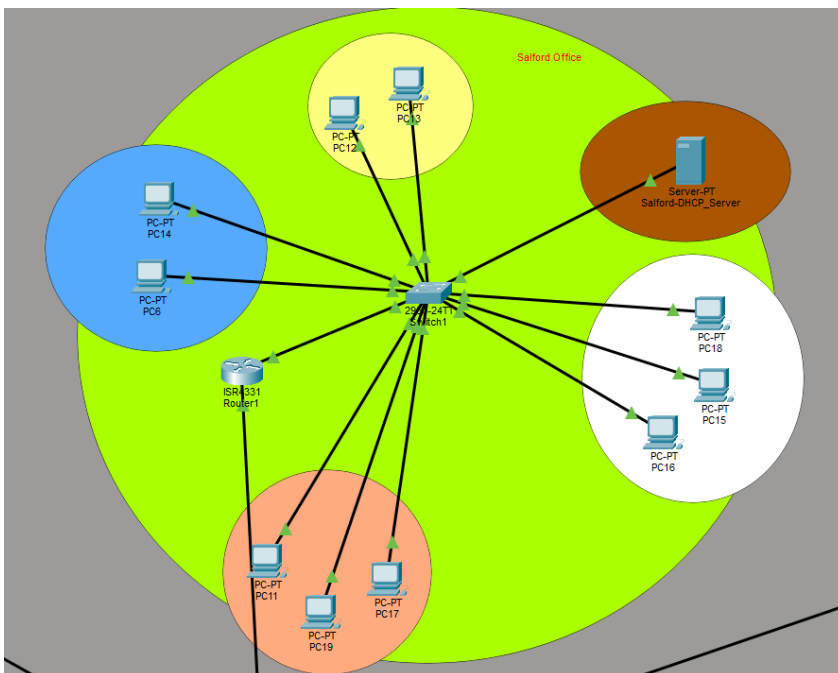
ALPHA-Logistics operates across four offices two in Manchester with a factory and two in London. The network was designed to support secure communications between departments, and provide reliable access to shared services such as DHCP, DNS, web and email. The network was designed to support future growth. Each site uses a star topology and pc connected to switches, VLANs provide department segmentation, and routers to provide inter-VLAN routing and WAN connectivity. The city of London offices act as a central branch/HQ providing access to core services and central point for inter-site routing



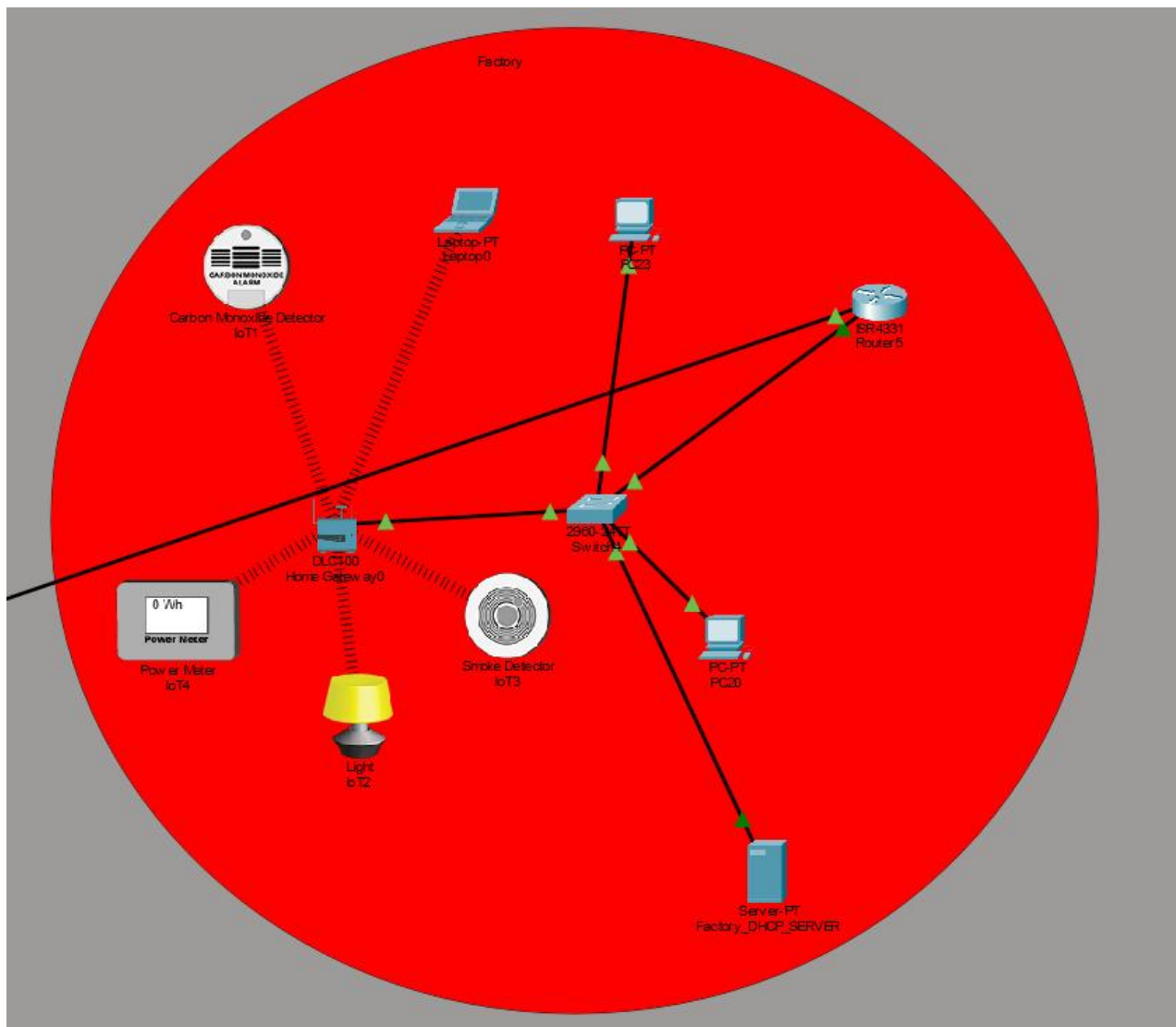
(Figure 1 – 4 yellow circle indicate office location with visually showing segmentation and 1 red circle indicate factory location)



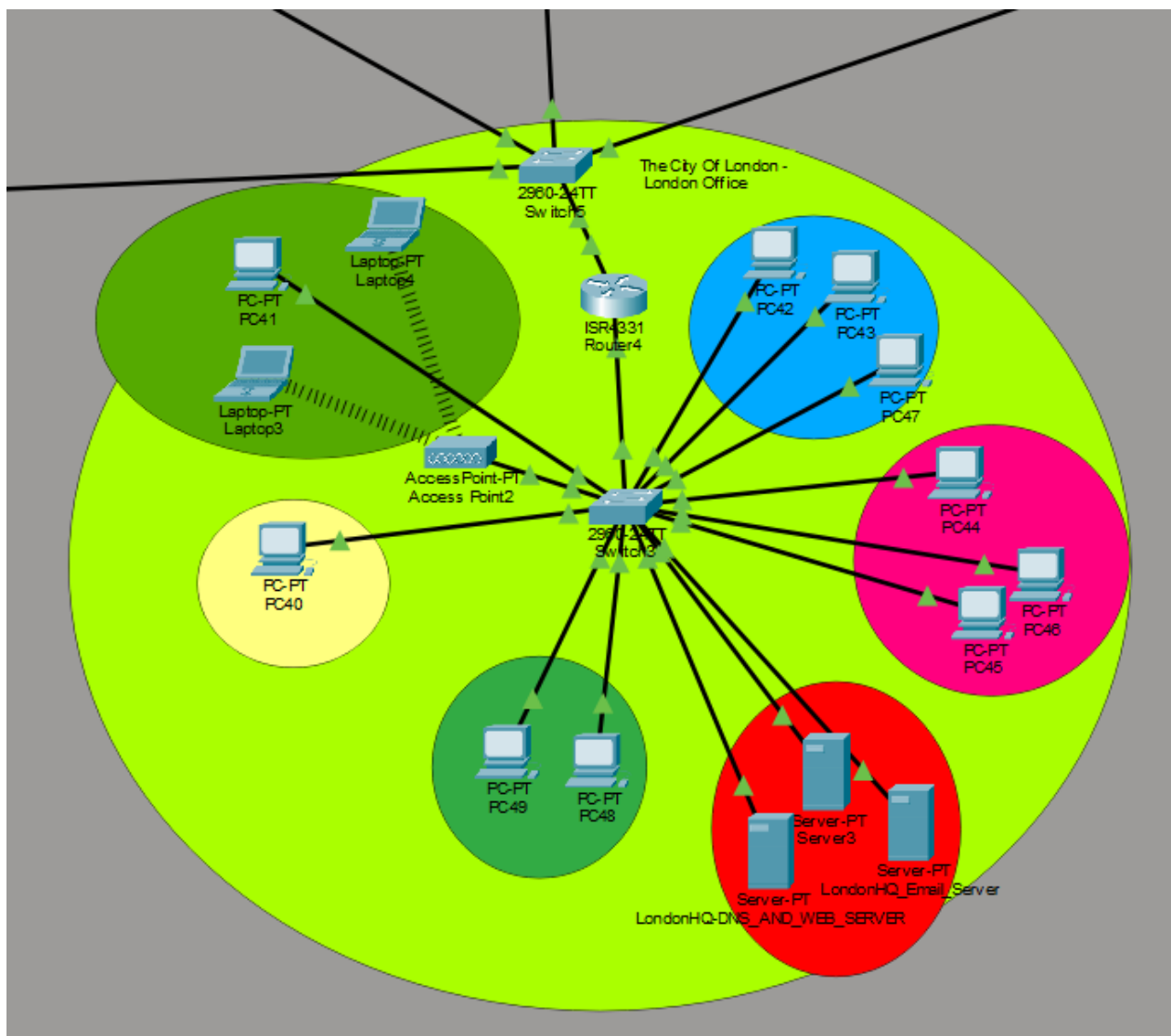
(Figure 2 – shows office in Manchester)



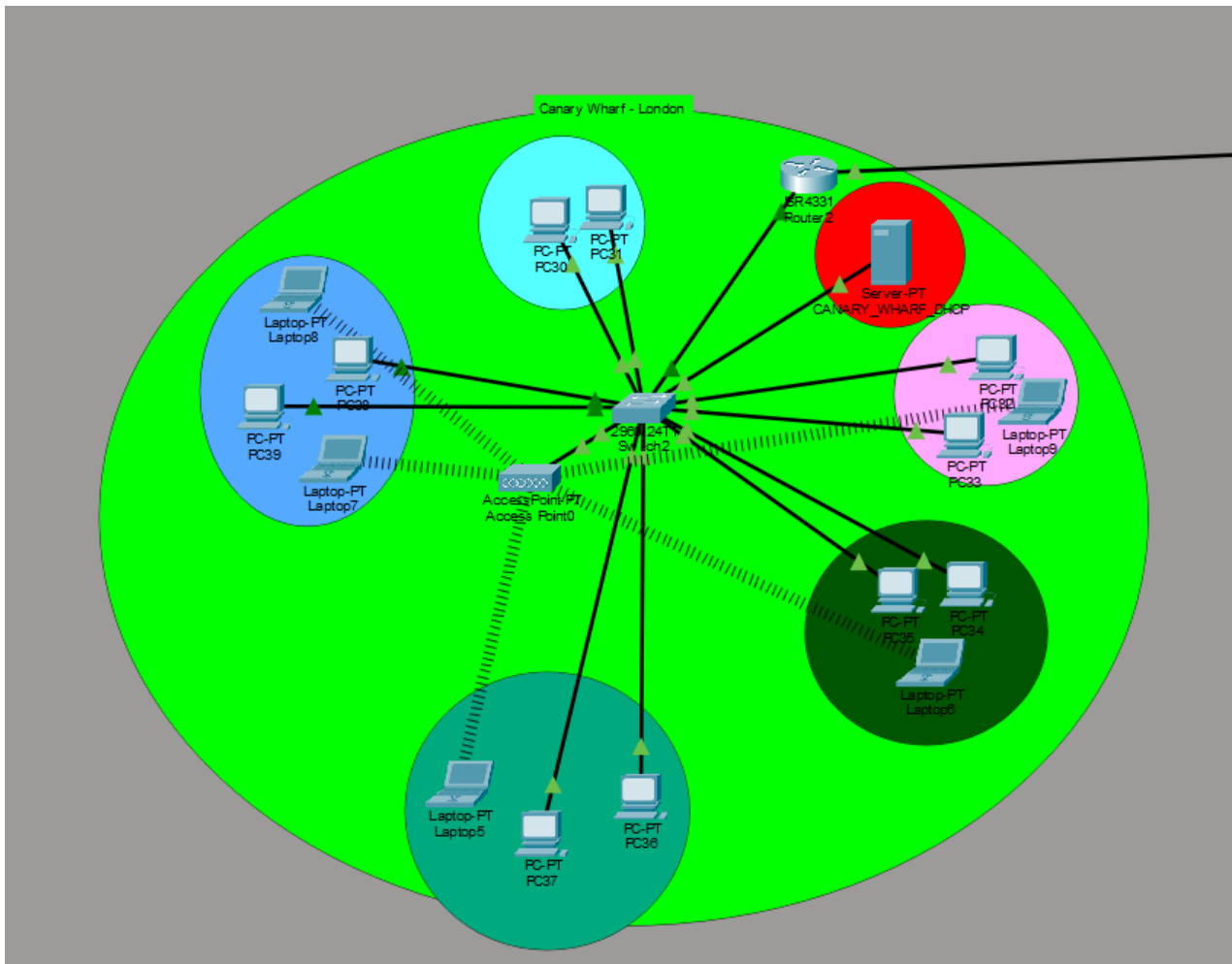
(Figure 3 – shows office in Manchester)



(Figure 4 – shows factory in Manchester)



(Figure 5 – shows office in London which is the main branch)



(Figure 6 – shows office in London)

Hardware and Topology

The design uses industry standard equipment available in packet tracer. Cisco 2960 switches provide VLAN-based segmentation and access connectivity. Cisco ISR4331 Routers provide inter-VLAN routing and inter-site routing. Each site includes PCs and at least one server. The employees also have an option to Bring their own devices such as laptops for wireless connection an access point is used to provide Wi-Fi access to laptops

A hub-and-spoke topology was selected with main branch at the centre. This simplifies routing and monitoring and provide centralised service management while keeping branch configuration consistent and providing scalability

```
SW-LondonHQ1#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Gig0/1, Gig0/2
10 IT_Department	active	Fa0/1, Fa0/12
20 Finance_Department	active	Fa0/2
30 EXECUTIVES_Department	active	Fa0/6, Fa0/7, Fa0/8
40 MarketingAndPR_Deparmtent	active	Fa0/3, Fa0/4
50 ReaserchAndDevelopment_Deparment	active	Fa0/5, Fa0/9, Fa0/10
89 SERVER_VLAN	active	Fa0/11, Fa0/13, Fa0/14
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW-LondonHQ1#
```

(Figure 7 – shows VLAN segmentation in City of London office)

```
LondonHQ#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0/0.10192.168.211.1		YES	manual	up	up
GigabitEthernet0/0/0.20192.168.221.1		YES	manual	up	up
GigabitEthernet0/0/0.30192.168.231.1		YES	manual	up	up
GigabitEthernet0/0/0.40192.168.241.1		YES	manual	up	up
GigabitEthernet0/0/0.50192.168.251.1		YES	manual	up	up
GigabitEthernet0/0/0.89192.168.249.1		YES	manual	up	up
GigabitEthernet0/0/1	10.0.0.1	YES	manual	up	up
GigabitEthernet0/0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
LondonHQ#
```

(Figure 8 – shows sub-interface segmentation in City of London office)

IP Addressing and Subnetting (CIDR)

The IP Addressing plan follows a structured approach to keep addressing clear and scalable. Each VLAN has its own subnet, and each site uses a unique address block to prevent overlap between London and Manchester networks. CIDR subnetting was used so address allocation matches the number of users per VLAN so that it reduces waste and support future expansion *VPC CIDR Blocks (2025)*.

For example, departments were assigned subnets such as /28 where the number of devices is small (up to 14 usable IPs) while larger segments can use larger masks such as /27 or /24. Router sub interfaces use the first usable IP (.1) in each subnet as the default gateway to maintain consistency and easy to troubleshoot

```

LondonHQ#show ip interface brief
Interface                IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0/0    unassigned      YES unset  up      up
GigabitEthernet0/0/0.10 192.168.211.1   YES manual up      up
GigabitEthernet0/0/0.20 192.168.221.1   YES manual up      up
GigabitEthernet0/0/0.30 192.168.231.1   YES manual up      up
GigabitEthernet0/0/0.40 192.168.241.1   YES manual up      up
GigabitEthernet0/0/0.50 192.168.251.1   YES manual up      up
GigabitEthernet0/0/0.89 192.168.249.1   YES manual up      up
GigabitEthernet0/0/1    10.0.0.1        YES manual up      up
GigabitEthernet0/0/2    unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
LondonHQ#

```

(Figure 9 – shows sub-interface segmentation in City of London office)

```

VLAN Name                Status    Ports
-----
1    default                active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Gig0/1
                                           Gig0/2
10   IT                      active    Fa0/1, Fa0/2
20   Sales_Department        active    Fa0/3, Fa0/4
30   HR_Department           active    Fa0/5, Fa0/12
40   Production_Department   active    Fa0/6, Fa0/8, Fa0/9
50   Finance_Department      active    Fa0/7, Fa0/10
89   SERVER_VLAN             active    Fa0/11
1002 fddi-default           active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
Switch#

```

(Figure 10 – shows VLAN segmentation in Manchester City Centre)

VLAN Segmentation Design

VLANs were implemented to separate departments and improve both performance and security. Without VLANs, all devices would share one broadcast domain increasing broadcast traffic and allowing unrestricted access between departments. With VLANs, each department becomes a separate logical network, and traffic can be controlled at layer 3 using the router *Network segmentation vs. Vlan explained (no date)* AlgoSec

Typical VLANs in this project include:

- IT
- Sales
- HR / Operations

- Production / Factory
- Server VLAN
- Staff Wi-Fi VLAN

VLAN	Name	Status	Ports
1	default	active	Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Gig0/1 Gig0/2
10	IT_Department	active	Fa0/1, Fa0/2
20	Legal_Department	active	Fa0/3, Fa0/4
30	HR_Department	active	Fa0/5, Fa0/6
40	SupplyChain_Mangement	active	Fa0/7, Fa0/8
50	Account_Mangement	active	Fa0/9, Fa0/10
60	STAFF_WiFi	active	Fa0/12
89	SERVER_VLAN	active	Fa0/11
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

(figure 10 – showing VLAN brief from Canary Wharf London Office)

Inter-VLAN Routing and Dynamic Routing - EIGRP

Inter-VLAN routing was configured as the end devices such as PCs and laptops in different VLANs can't communicate by default. A single physical router interface connects to the switch using a trunk link, carrying multiple VLANs using 802.1Q tagging. The router uses sub interfaces such as G0/0/0.10,0.20,0.30 where each sub interface shows a VLAN gateway. *Introduction to eigrp (2025) Cisco*

This approach reduces the hardware requirements compared to using a separate physical router port per VLAN while still providing full layer 3 connectivity and control between VLANs.

To enable communication between sites, dynamic routing was implemented using EIGRP as you can see in figure 10 which automatically exchanges routing information between routers, so routes do not need to be manually entered for each new subnet which improves scalability and reduces configuration errors as the network grows

I have decided to use EIGRP because it provides fast coverage and works well for multi-site network designs. The WAN addressing between routers uses a separate network to clearly show transit links from LAN VLANs. Successful neighbour confirms that routers can exchange routing updates.

```

LondonHQ#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/24 is directly connected, GigabitEthernet0/0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
D    192.168.10.0/28 is subnetted, 1 subnets
     192.168.10.0/28 [90/28416] via 10.0.0.2, 00:07:27, GigabitEthernet0/0/1
D    192.168.20.0/28 is subnetted, 1 subnets
     192.168.20.0/28 [90/28416] via 10.0.0.2, 00:07:27, GigabitEthernet0/0/1
D    192.168.30.0/28 is subnetted, 1 subnets
     192.168.30.0/28 [90/28416] via 10.0.0.2, 00:07:27, GigabitEthernet0/0/1
D    192.168.40.0/28 is subnetted, 1 subnets
     192.168.40.0/28 [90/28416] via 10.0.0.2, 00:07:27, GigabitEthernet0/0/1
D    192.168.50.0/28 is subnetted, 1 subnets
     192.168.50.0/28 [90/28416] via 10.0.0.2, 00:07:27, GigabitEthernet0/0/1
D    192.168.60.0/28 is subnetted, 1 subnets
     192.168.60.0/28 [90/28416] via 10.0.0.4, 00:07:27, GigabitEthernet0/0/1
D    192.168.70.0/28 is subnetted, 1 subnets
     192.168.70.0/28 [90/28416] via 10.0.0.4, 00:07:27, GigabitEthernet0/0/1
D    192.168.80.0/28 is subnetted, 1 subnets
     192.168.80.0/28 [90/28416] via 10.0.0.4, 00:07:27, GigabitEthernet0/0/1
D    192.168.89.0/28 is subnetted, 1 subnets
     192.168.89.0/28 [90/28416] via 10.0.0.2, 00:07:27, GigabitEthernet0/0/1
D    192.168.110.0/28 is subnetted, 1 subnets
     192.168.110.0/28 [90/28416] via 10.0.0.3, 00:07:27, GigabitEthernet0/0/1
D    192.168.120.0/28 is subnetted, 1 subnets
     192.168.120.0/28 [90/28416] via 10.0.0.3, 00:07:27, GigabitEthernet0/0/1
D    192.168.130.0/28 is subnetted, 1 subnets
     192.168.130.0/28 [90/28416] via 10.0.0.3, 00:07:27, GigabitEthernet0/0/1
D    192.168.140.0/28 is subnetted, 1 subnets
     192.168.140.0/28 [90/28416] via 10.0.0.3, 00:07:27, GigabitEthernet0/0/1
D    192.168.150.0/28 is subnetted, 1 subnets
     192.168.150.0/28 [90/28416] via 10.0.0.3, 00:07:27, GigabitEthernet0/0/1
D    192.168.160.0/28 is subnetted, 1 subnets
     192.168.160.0/28 [90/28416] via 10.0.0.5, 00:07:27, GigabitEthernet0/0/1
D    192.168.161.0/28 is subnetted, 1 subnets
     192.168.161.0/28 [90/28416] via 10.0.0.5, 00:07:27, GigabitEthernet0/0/1
D    192.168.170.0/28 is subnetted, 1 subnets
     192.168.170.0/28 [90/28416] via 10.0.0.5, 00:07:27, GigabitEthernet0/0/1
D    192.168.180.0/28 is subnetted, 1 subnets
     192.168.180.0/28 [90/28416] via 10.0.0.5, 00:07:27, GigabitEthernet0/0/1
D    192.168.189.0/28 is subnetted, 1 subnets
     192.168.189.0/28 [90/28416] via 10.0.0.3, 00:07:27, GigabitEthernet0/0/1
D    192.168.190.0/28 is subnetted, 1 subnets
     192.168.190.0/28 [90/28416] via 10.0.0.5, 00:07:27, GigabitEthernet0/0/1
D    192.168.200.0/28 is subnetted, 1 subnets
     192.168.200.0/28 [90/28416] via 10.0.0.5, 00:07:27, GigabitEthernet0/0/1
D    192.168.209.0/28 is subnetted, 1 subnets
     192.168.209.0/28 [90/28416] via 10.0.0.5, 00:07:27, GigabitEthernet0/0/1
D    192.168.211.0/24 is variably subnetted, 2 subnets, 2 masks

LondonHQ#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address           Interface      Hold Uptime    SRTT    RTO    Q   Seq
                               (sec)          (ms)                Cnt   Num
0   10.0.0.3           Gig0/0/1       10   00:12:31     40     1000    0    48
1   10.0.0.5           Gig0/0/1       12   00:12:31     40     1000    0    46
2   10.0.0.2           Gig0/0/1       12   00:12:31     40     1000    0    48
3   10.0.0.4           Gig0/0/1       14   00:12:31     40     1000    0    54

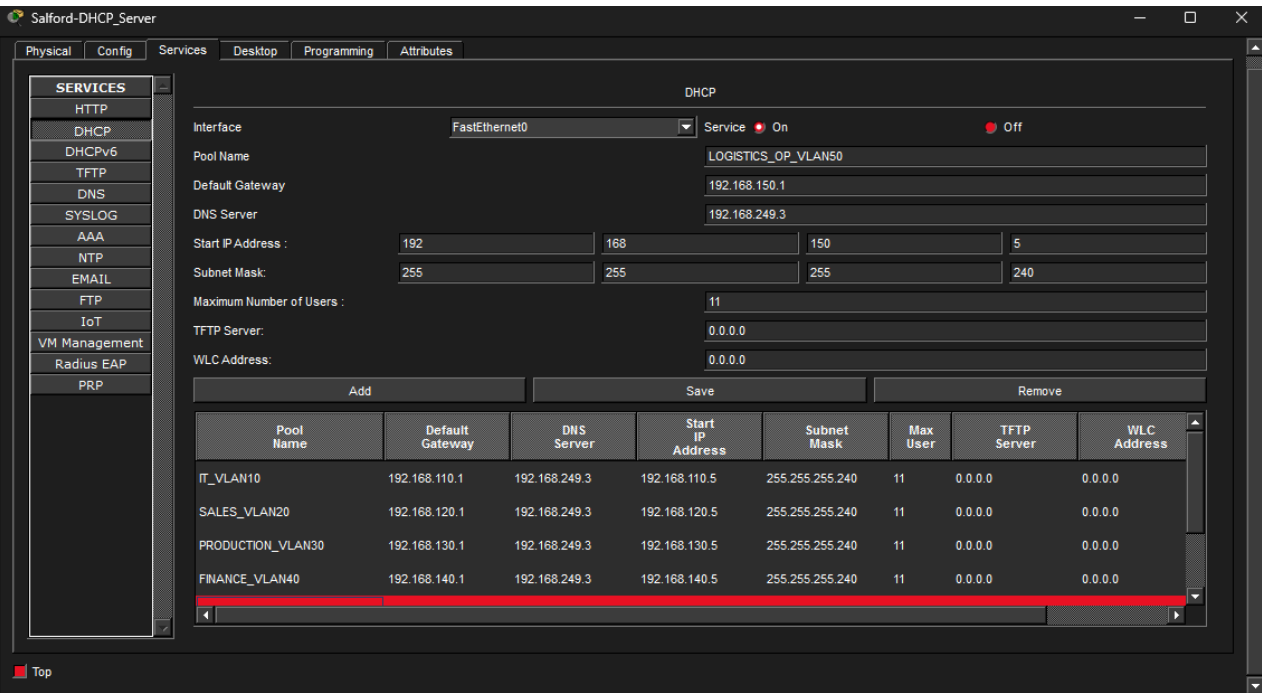
LondonHQ#

```

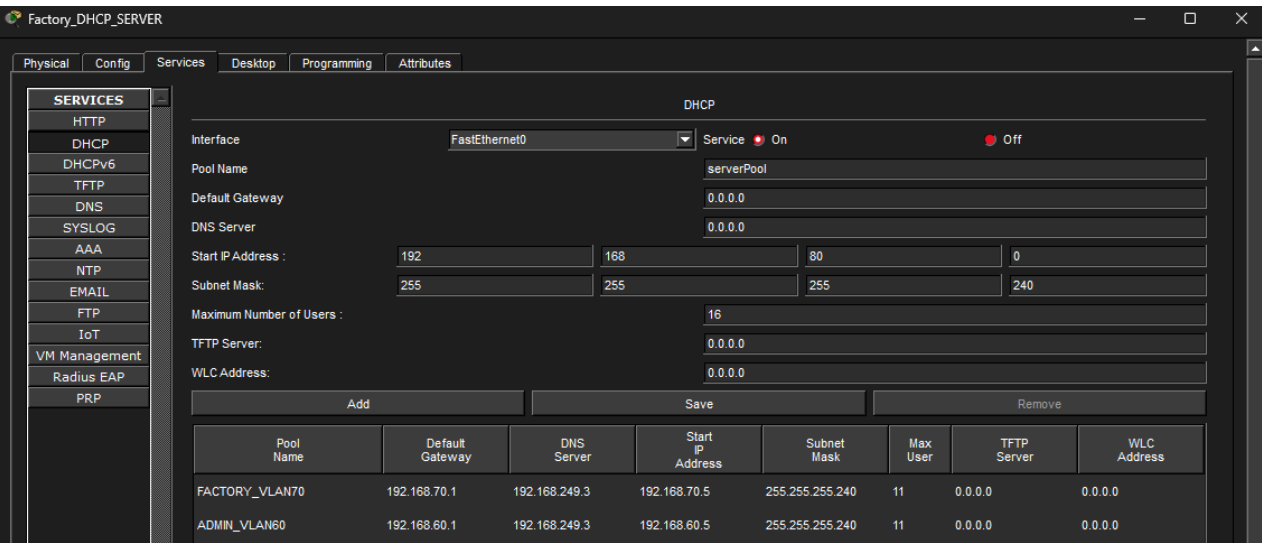
(Figure 10 and 11 – shows EIGRP is working correctly and LondonHQ (City of London) has learned all remote office subnets)

DHCP design

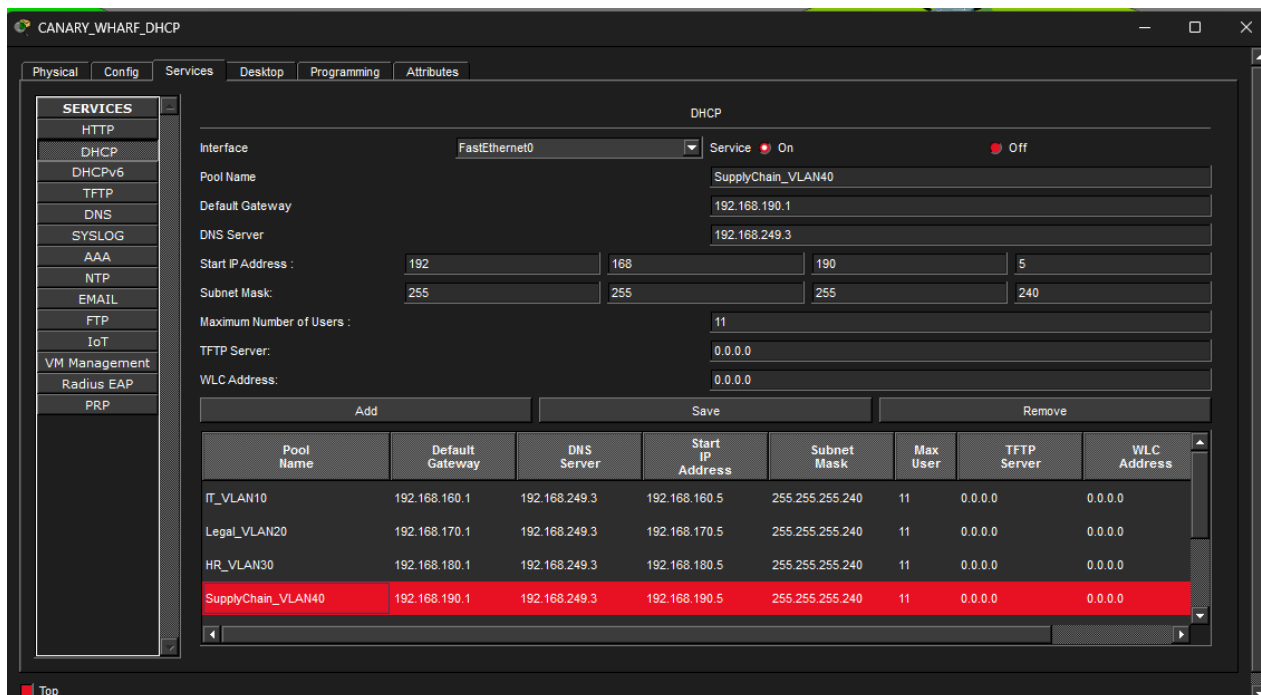
I have implemented DHCP to automatically provide IP configuration like IP address, subnet mask, gateway, DNS server to clients in each VLAN. This reduces manual configuration effort and prevents any addressing mistakes. Because the DHCP server is not physically inside every VLAN, I have configured routers to forward DHCP request using DHCP relay. Each VLAN has its own DHCP pool matching its subnet and gateway. This ensures end devices always receive address appropriate to their VLAN and can route correctly.



(figure 12 – showing Salford DHCP server)



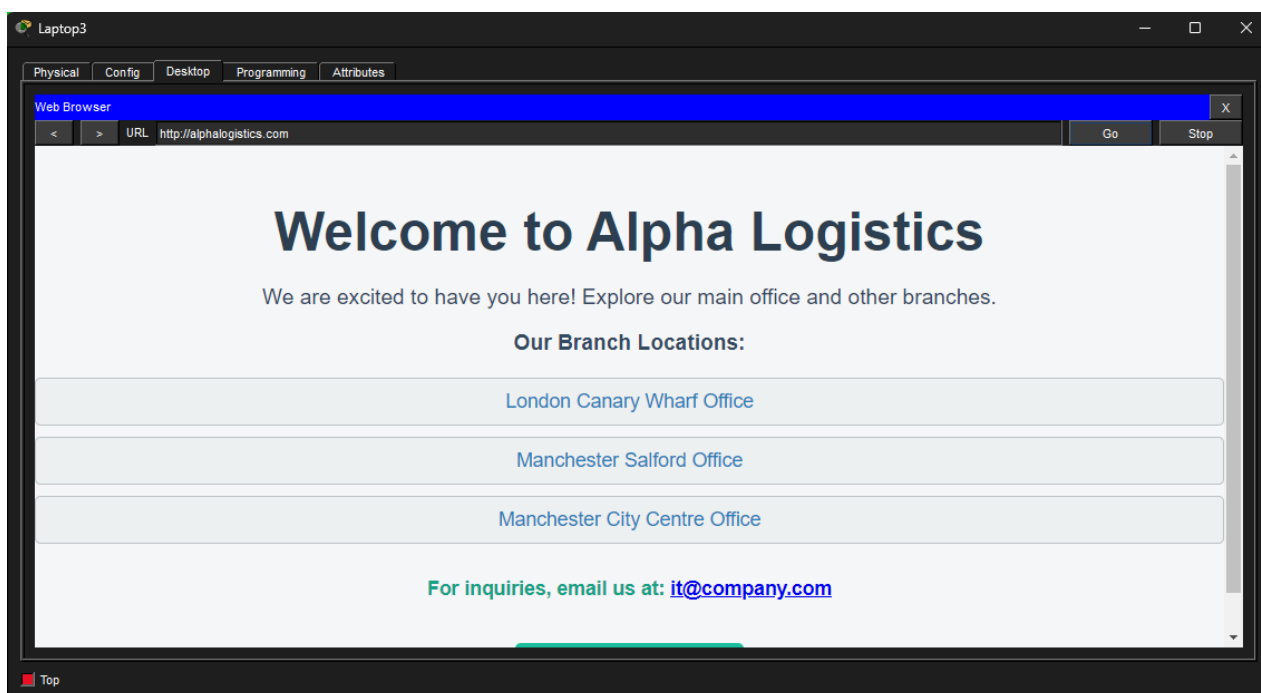
(figure 13 – showing Factory DHCP server)



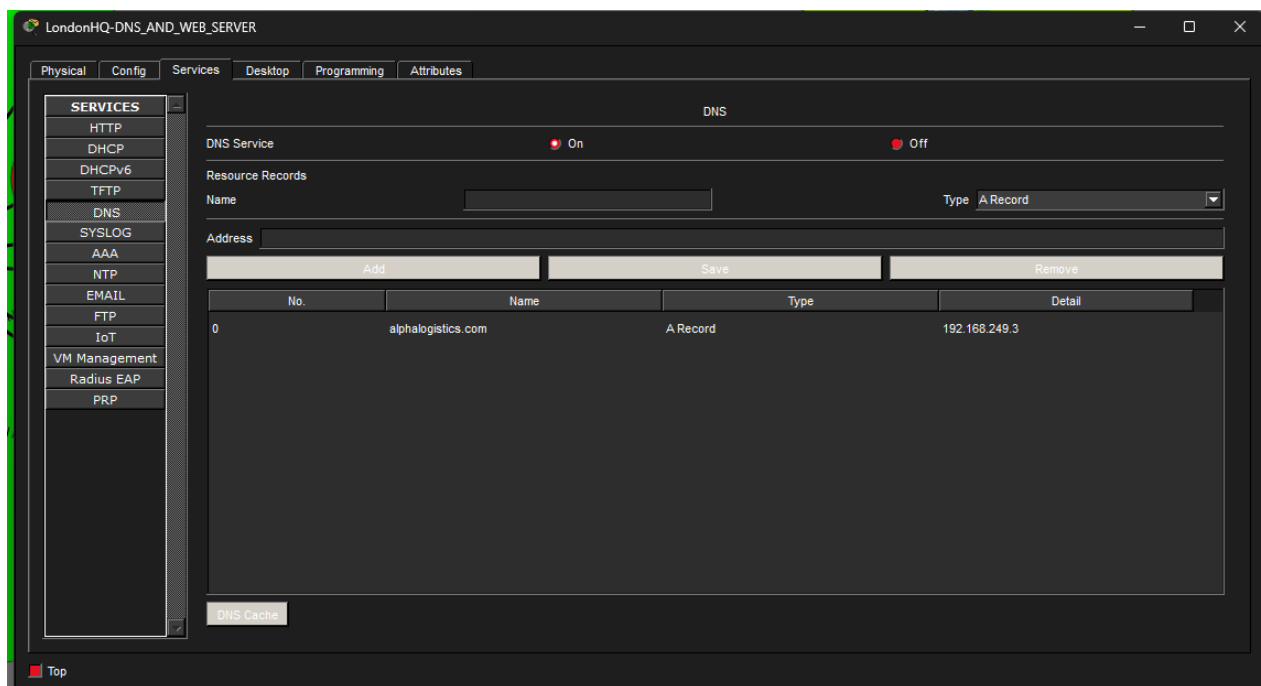
(figure 14 – showing Canary Wharf DHCP server)

DNS and Web service Design

I have implemented DNS service so users can access services using names rather than IP addresses. For example, alphalogistics.com was mapped to the internal web server IP. DHCP was configured to provide the DNS server address automatically to all clients, so name resolution works on all networks. The HTTP server hosts the company website and allows access from any VLAN and any office, demonstrating effective end-to-end connectivity and application layer functionality

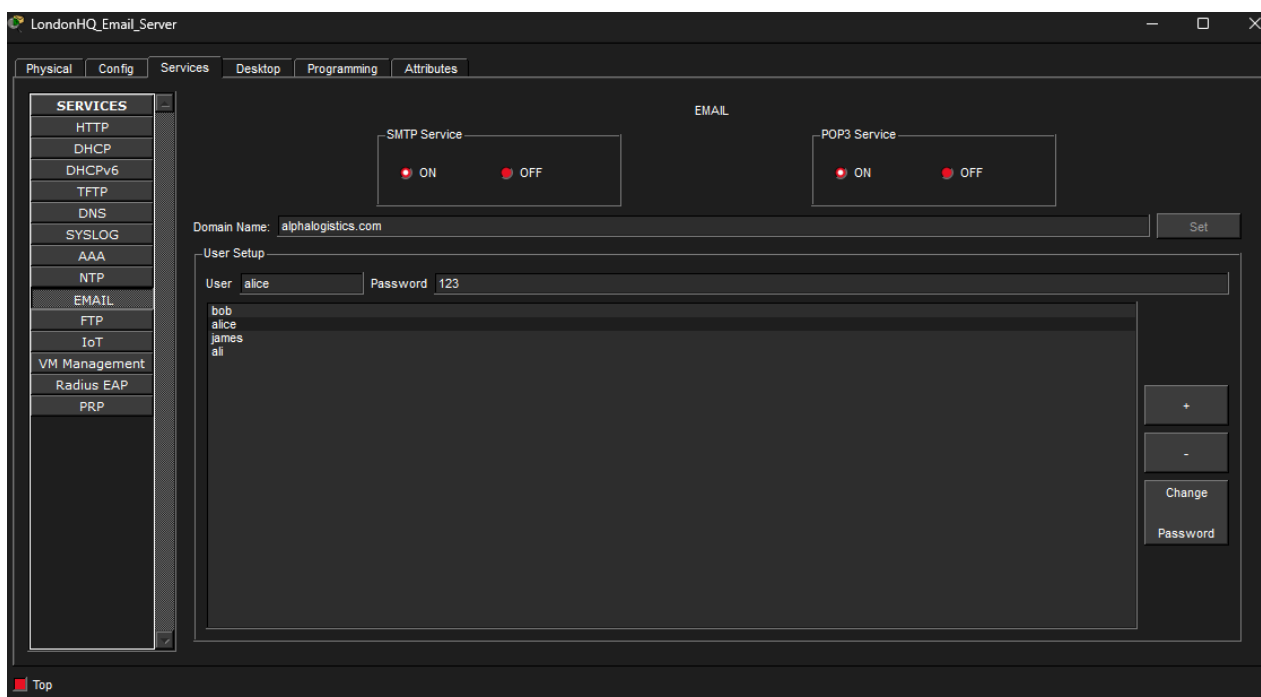


(figure 15 – showing website is correctly loaded from DNS)



Email Service Design

I have implemented and configured an email server in main branch of London office to support internal communication across all locations; User Accounts were created on mail servers and clients were configured with the correct mail server address and DNS settings. By using a central domain and ensuring routing and DNS are correct, email communication works across Manchester and London, showing inter-site routing and service reachability



(figure 16 – shows email accounts configured to send and receive emails)

Wireless Network Design (Staff Wi-Fi VLAN)

Wireless access was added for staff laptops using a dedicated Wi-Fi VLAN. This allows wireless devices are separated from internal wired departments while still being able to reach shared services through routing. This was done so that policy like BYOD which carries significant threat of spreading malware to internal networks to be separated and improve security. The access point is connected to the switch and switch port is configured to place wireless traffic into the correct VLAN. DHCP provides addresses to wireless clients to access internal services.

```
C:\>ping 192.168.40.5

Pinging 192.168.40.5 with 32 bytes of data:

Request timed out.
Reply from 192.168.40.5: bytes=32 time=36ms TTL=126
Reply from 192.168.40.5: bytes=32 time=27ms TTL=126
Reply from 192.168.40.5: bytes=32 time=22ms TTL=126

Ping statistics for 192.168.40.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 36ms, Average = 28ms

C:\>ipconfig /all

Wireless0 Connection: (default port)

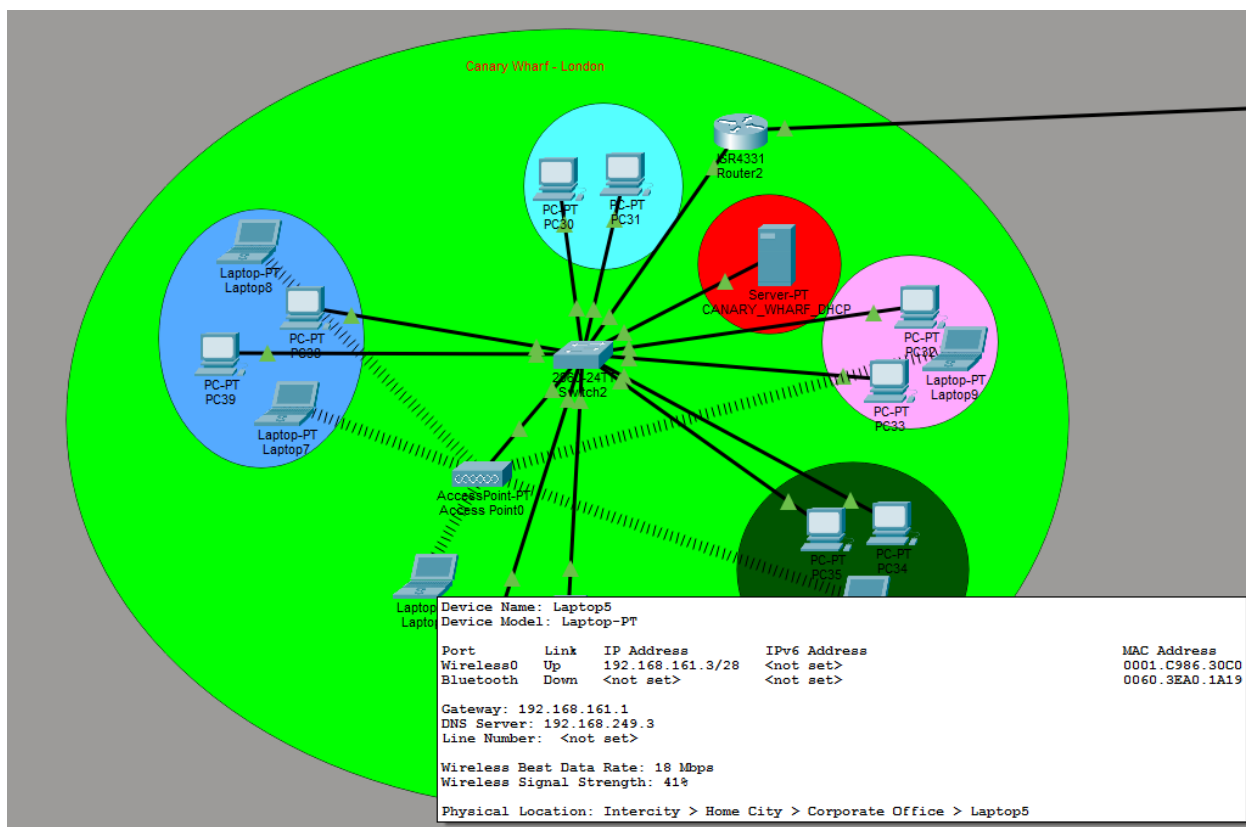
    Connection-specific DNS Suffix...: 
    Physical Address. . . . . : 0001.C986.30C0
    Link-local IPv6 Address . . . . . : FE80::201:C9FF:FE86:30C0
    IPv6 Address. . . . . : ::
    IPv4 Address. . . . . : 192.168.161.3
    Subnet Mask. . . . . : 255.255.255.240
    Default Gateway. . . . . : ::
                                192.168.161.1
    DHCP Servers. . . . . : 192.168.209.10
    DHCPv6 IAID. . . . . : 1255879017
    DHCPv6 Client DUID. . . . . : 00-01-00-01-9A-8A-32-93-00-01-C9-86-30-C0
    DNS Servers. . . . . : ::
                                192.168.249.3

Bluetooth Connection:

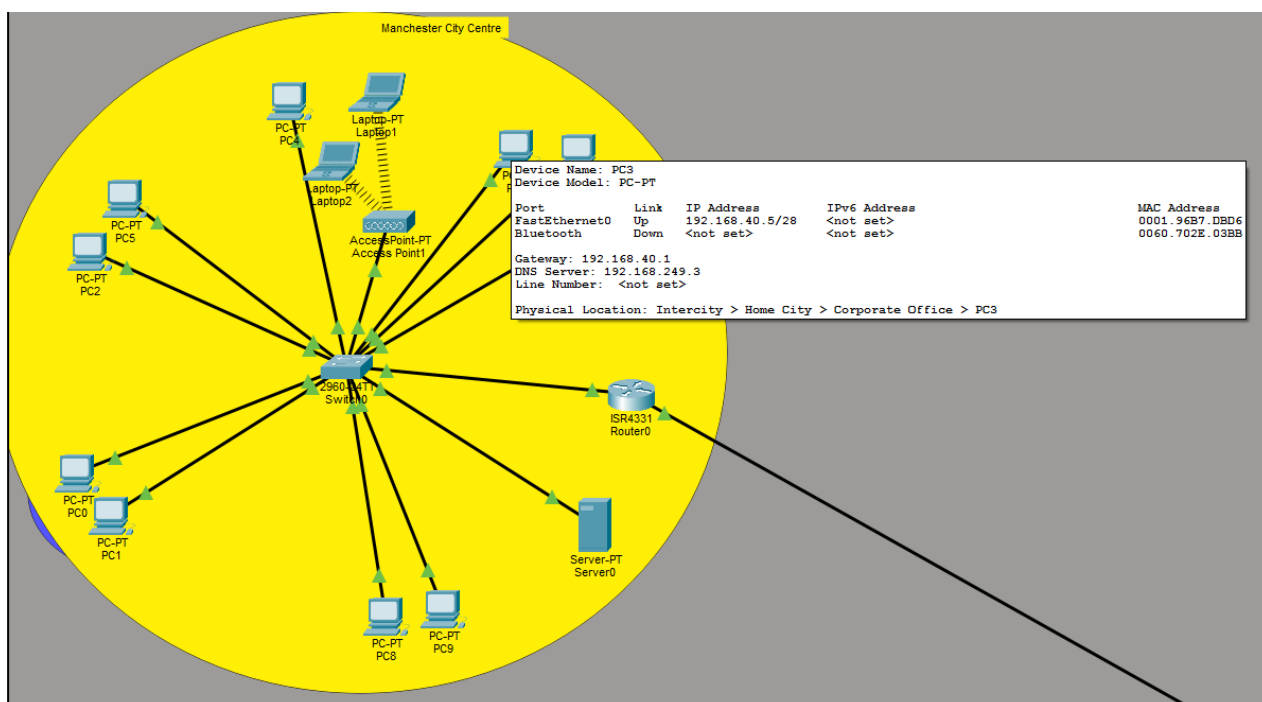
    Connection-specific DNS Suffix...: 
    Physical Address. . . . . : 0060.3EA0.1A19
    Link-local IPv6 Address . . . . . : ::
    IPv6 Address. . . . . : ::
    IPv4 Address. . . . . : 0.0.0.0
    Subnet Mask. . . . . : 0.0.0.0
    Default Gateway. . . . . : ::
                                0.0.0.0
    DHCP Servers. . . . . : 0.0.0.0
    DHCPv6 IAID. . . . . : 1255879017
    DHCPv6 Client DUID. . . . . : 00-01-00-01-9A-8A-32-93-00-01-C9-86-30-C0
    DNS Servers. . . . . : ::
                                192.168.249.3

C:\>
```

(figure 17 – shows wireless device successfully communicating with internal network)



(figure 18 – Evidence for wireless device with its IP address)



(figure 19 – Evidence for wired device with its IP address)

Security Design

Infrastructure devices were hardened to prevent unauthorized access and reduce attack surface. All routers and switches were configured with secure passwords, local admin accounts, encrypted passwords, and SSH for remote management, insecure services were disabled when not required. On switches and routers passwords were implemented and unused ports were shut down to prevent unauthorised devices

```
User Access Verification

Password:

LondonHQ>enable
Password:
LondonHQ#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
LondonHQ#
```

(figure 20 – shows ssh was configured for City of London router)

```
User Access Verification

Username: admin
Password:

SalfordOffice_Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
SalfordOffice_Router#
```

```
Device Name: Switch0
Custom Device Model: 2960 IOS15
Hostname: MANCHESTER_CC_SWITCH
```

Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	10	--	00D0.BC95.2501
FastEthernet0/2	Up	10	--	00D0.BC95.2502
FastEthernet0/3	Up	20	--	00D0.BC95.2503
FastEthernet0/4	Up	20	--	00D0.BC95.2504
FastEthernet0/5	Up	30	--	00D0.BC95.2505
FastEthernet0/6	Up	40	--	00D0.BC95.2506
FastEthernet0/7	Up	50	--	00D0.BC95.2507
FastEthernet0/8	Up	40	--	00D0.BC95.2508
FastEthernet0/9	Up	40	--	00D0.BC95.2509
FastEthernet0/10	Up	50	--	00D0.BC95.250A
FastEthernet0/11	Up	89	--	00D0.BC95.250B
FastEthernet0/12	Up	30	--	00D0.BC95.250C
FastEthernet0/13	Down	1	--	00D0.BC95.250D
FastEthernet0/14	Down	1	--	00D0.BC95.250E
FastEthernet0/15	Down	1	--	00D0.BC95.250F
FastEthernet0/16	Down	1	--	00D0.BC95.2510
FastEthernet0/17	Down	1	--	00D0.BC95.2511
FastEthernet0/18	Down	1	--	00D0.BC95.2512
FastEthernet0/19	Down	1	--	00D0.BC95.2513
FastEthernet0/20	Down	1	--	00D0.BC95.2514
FastEthernet0/21	Down	1	--	00D0.BC95.2515
FastEthernet0/22	Down	1	--	00D0.BC95.2516
FastEthernet0/23	Down	1	--	00D0.BC95.2517
FastEthernet0/24	Up	--	--	00D0.BC95.2518
GigabitEthernet0/1	Down	1	--	00D0.BC95.2519
GigabitEthernet0/2	Down	1	--	00D0.BC95.251A
Vlan1	Down	1	<not set>	00D0.FF69.5385

(figure 22 – shows shutting down unused ports for security purposes)

Test ID	Description	Source	Destination	Expected Result	Actual Result	Status
T00	Inter-VLAN communication Test in Manchester City Centre Office	PC1 – 192.168.10.6	Laptop1 – 192.168.30.5	PC1 should be able to ping laptop1	Confirms Inter-VLAN communication works	Pass
T01	Inter-VLAN communication Test in City of London office	PC41 – 192.168.211.7	PC48 – 192.168.241.5	PC41 should be able to ping PC48	Confirms Inter-VLAN communication works	Pass
T02	Inter-site connectivity from Canary Wharf to Salford	PC39 – 192.168.170.5	PC15 – 192.168.150.6	PC39 should be able to ping PC15 which in Salford office	Confirms inter-site connectivity is functional	Pass
T03	Inter-site connectivity from the City of London to Factory	PC46 – 192.168.231.7	PC20 – 192.168.60.5	PC46 should be able to ping PC20	Confirms inter-site connectivity is functional	Pass
T04	DHCP server test in the city of London	N/A	N/A	DHCP server pool should give out IP address	DHCP server is functional in the City of London office	Pass
T05	DHCP server test in factory	N/A	N/A	DHCP server pool should give out IP address	DHCP server is functional in factory	Pass
T06	Wireless Connectivity Test	Laptop2 – 192.168.30.7	Laptop9 – 192.168.161.7	Laptop2 should be able to ping laptop9	Shows wireless connectivity works	Pass
T07	DNS and Web server – Factory location	PC20	www.alphalogistics.com	PC20 should be able to access company website	DNS and Web were accessible from the pc showing it was functional	Pass
T08	DNS and Web Server – Manchester City Centre Office	PC8	www.alphalogistics.com	PC8 should be able to access company website	DNS and Web were accessible from the pc showing it was functional	Pass
T09	Email Service test from Factory to Canary Wharf	PC23 – 192.168.60.6 - tony@alphalogistics.com	Laptop7 – 192.168.161.6 tyler@alphalogistics.com	Email should be sent and received	Shows email service is functional	Pass
T10	Email Service test from Manchester	PC2- 192.168.20.5 – bob@alphalogistics.com	PC18 – 192.168.150.7 – James@alphalogistics.com			

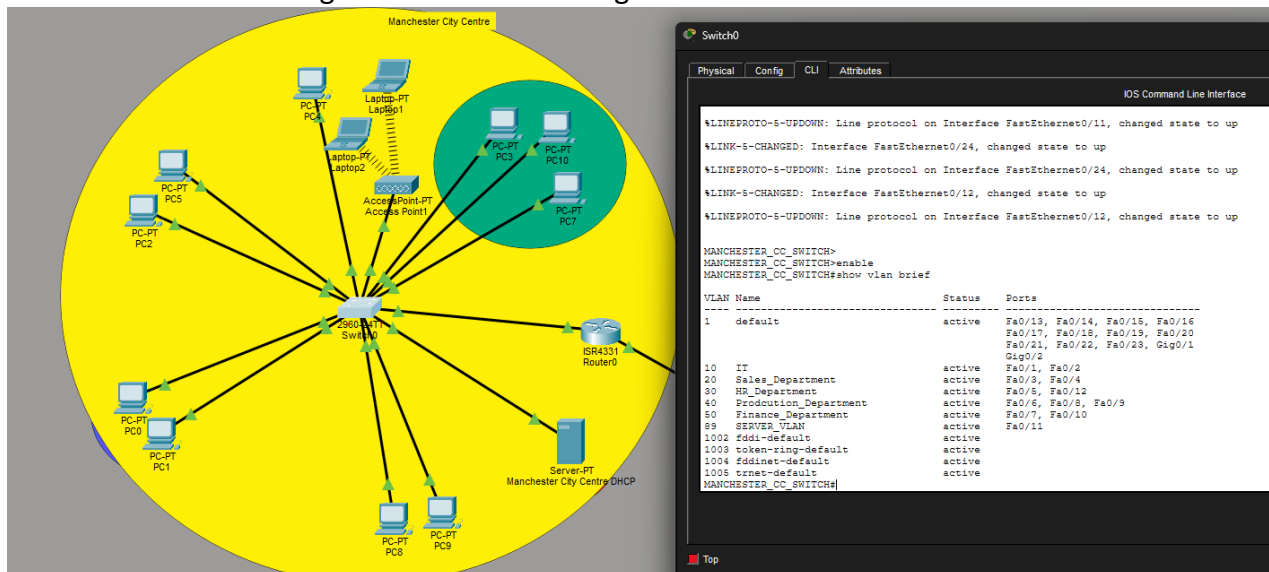
	City Centre to Salford					
T11	Router Configuration test from Salford to Factory	10.0.0.3	10.0.0.4	Should be able to ping each other	Worked as expected	Pass
T12	Router Configuration Test from City of London to Manchester City Centre	10.0.0.2	10.0.0.1	Should be able to ping each other	Worked as expected	Pass

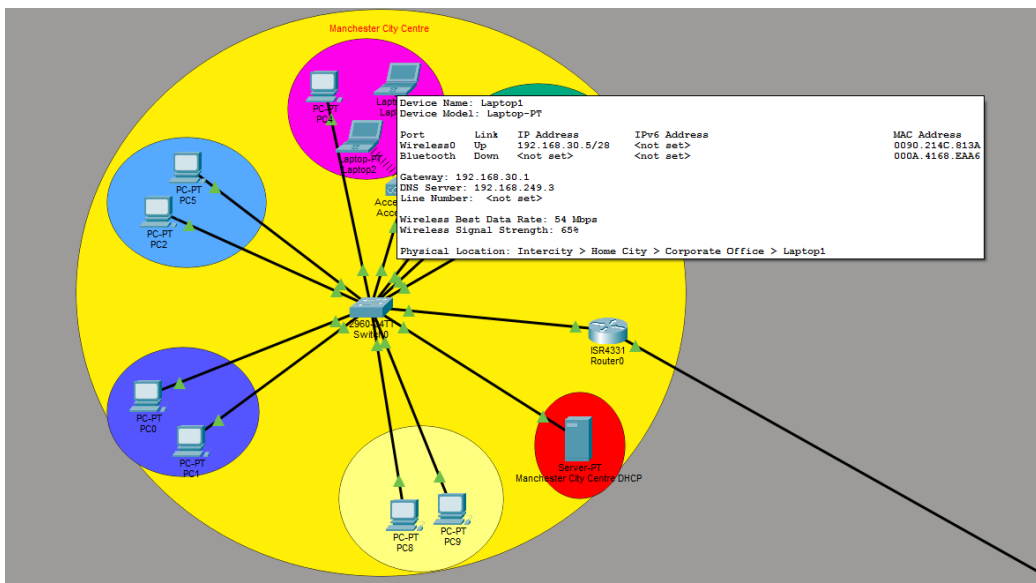
Network Configuration Tests

Inter-VLAN communication Test

T00 - Manchester City Centre Office

This test shows VLAN segmentation and routing works





```

PC1

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address...: 0001.C720.CB54
Link-local IPv6 Address...: FE80::201:C7FF:FE20:CB54
IPv6 Address...: ::
IPv4 Address...: 192.168.10.6
Subnet Mask...: 255.255.255.240
Default Gateway...: ::
192.168.10.1
DHCP Servers...: 192.168.89.10
DHCPv6 IAID...:
DHCPv6 Client DUID...: 00-01-00-01-37-A9-91-4E-00-01-C7-20-CB-54
DNS Servers...: ::
192.168.249.3

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address...: 0001.96D4.CB3A
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: ::
0.0.0.0
DHCP Servers...: 0.0.0.0
DHCPv6 IAID...:
DHCPv6 Client DUID...: 00-01-00-01-37-A9-91-4E-00-01-C7-20-CB-54
DNS Servers...: ::
192.168.249.3

C:\>ping 192.168.30.5

Pinging 192.168.30.5 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.5: bytes=32 time=23ms TTL=127
Reply from 192.168.30.5: bytes=32 time=10ms TTL=127
Reply from 192.168.30.5: bytes=32 time=16ms TTL=127

Ping statistics for 192.168.30.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 23ms, Average = 16ms

C:\>ping 192.168.30.5

Pinging 192.168.30.5 with 32 bytes of data:

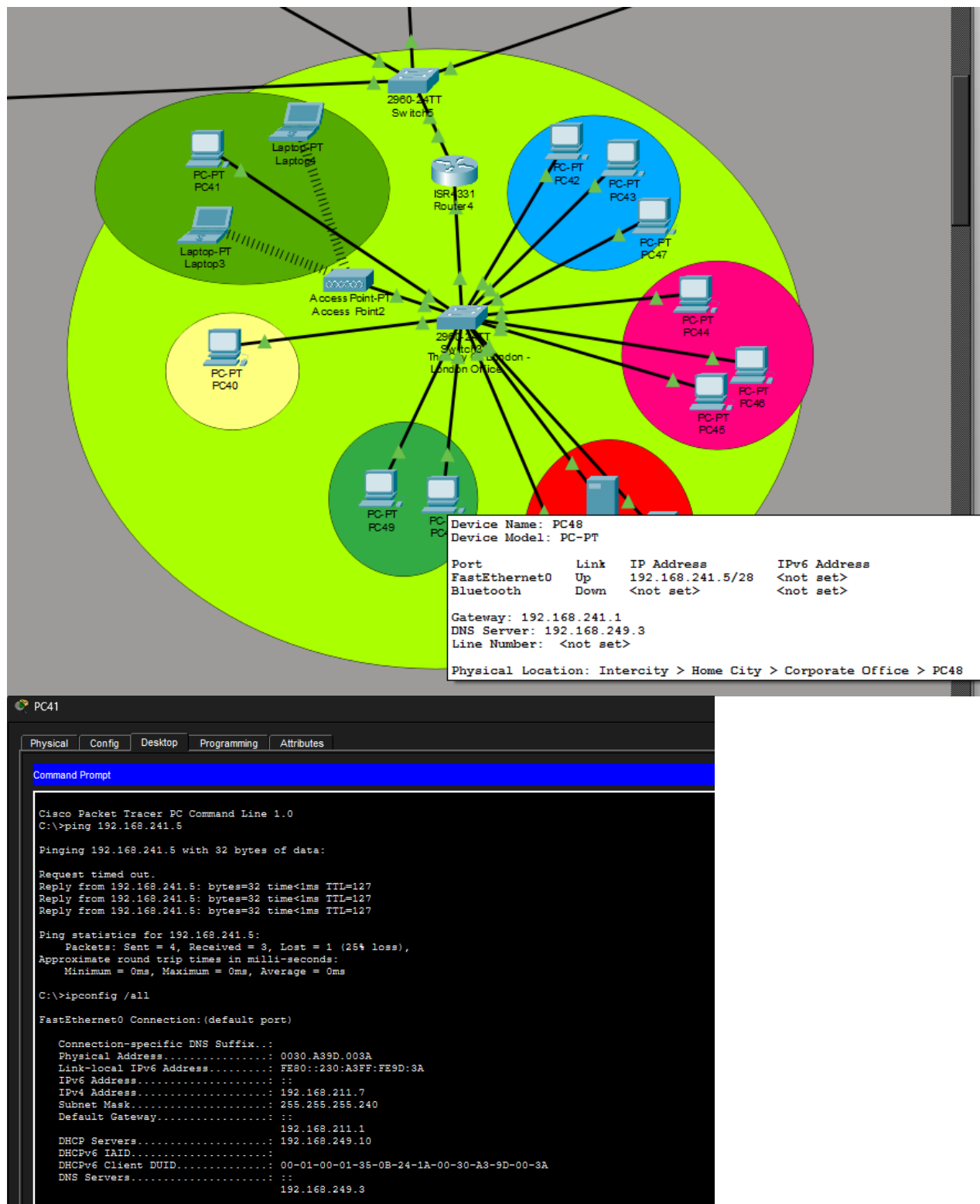
Reply from 192.168.30.5: bytes=32 time=16ms TTL=127
Reply from 192.168.30.5: bytes=32 time=21ms TTL=127
Reply from 192.168.30.5: bytes=32 time=16ms TTL=127
Reply from 192.168.30.5: bytes=32 time=16ms TTL=127

Ping statistics for 192.168.30.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 21ms, Average = 17ms

C:\>|
  
```

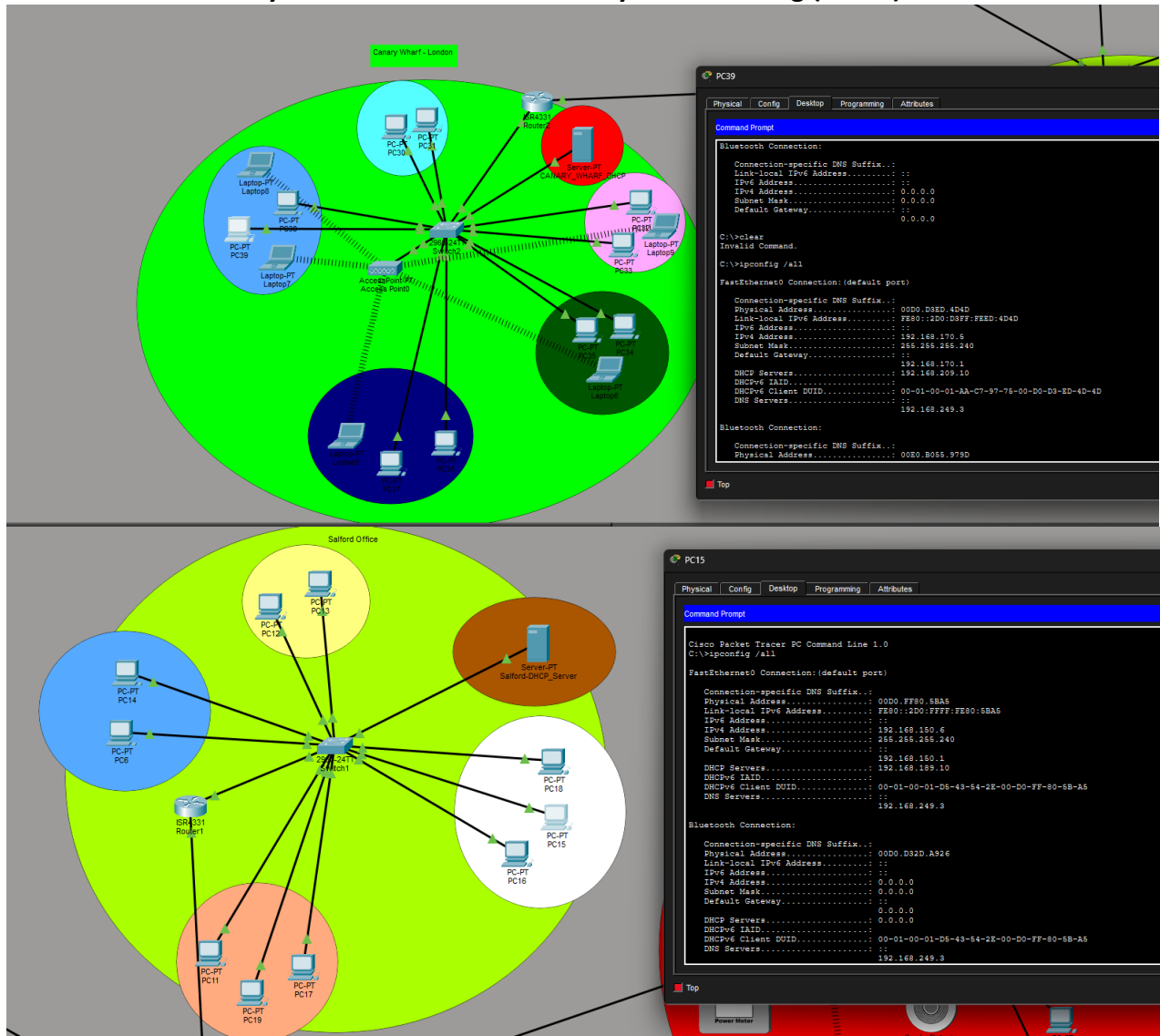
T01 - The City of London Office

This test shows if VLAN segmentation and routing works



Inter-site connectivity Test

T02 - Test from Canary Wharf to Salford to see if Dynamic Routing (EIGRP) works




```
PC39
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig /all
Ethernet0 Connection: (default port)
    Connection-specific DNS Suffix...: ::
    Default Gateway...: ::
    192.168.170.1

Bluetooth Connection:
    Connection-specific DNS Suffix...: ::
    Link-local IPv6 Address...: ::
    IPv6 Address...: ::
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: ::
    0.0.0.0

C:\>clear
Invalid Command.

C:\>ipconfig /all
FastEthernet0 Connection: (default port)
    Connection-specific DNS Suffix...: ::
    Physical Address...: 00D0.D3ED.4D4D
    Link-local IPv6 Address...: FE80::2D0:D3FF:FEED:4D4D
    IPv6 Address...: ::
    IPv4 Address...: 192.168.170.5
    Subnet Mask...: 255.255.255.240
    Default Gateway...: ::
    192.168.170.1
    DHCP Servers...: 192.168.209.10
    DHCPv6 IAID...:
    DHCPv6 Client DUID...: 00-01-00-01-AA-C7-97-75-00-D0-D3-ED-4D-4D
    DNS Servers...: ::
    192.168.249.3

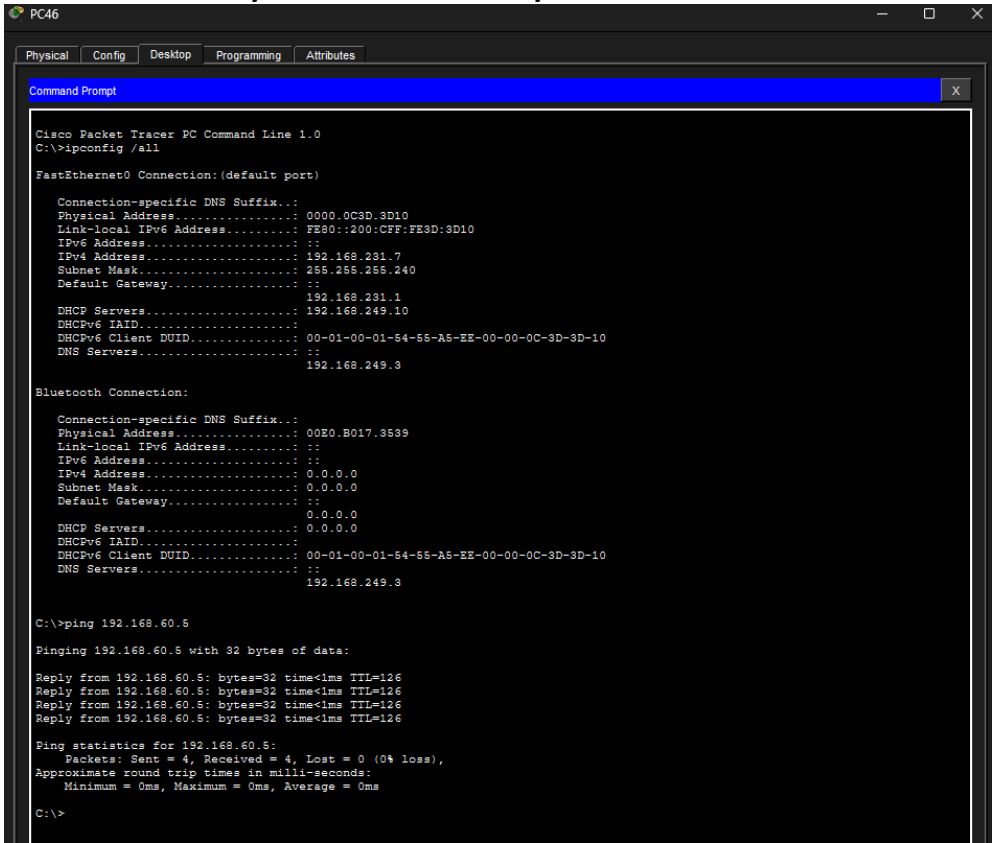
Bluetooth Connection:
    Connection-specific DNS Suffix...: ::
    Physical Address...: 00E0.B055.979D
    Link-local IPv6 Address...: ::
    IPv6 Address...: ::
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: ::
    0.0.0.0
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...:
    DHCPv6 Client DUID...: 00-01-00-01-AA-C7-97-75-00-D0-D3-ED-4D-4D
    DNS Servers...: ::
    192.168.249.3

C:\>ping 192.168.160.6
Pinging 192.168.160.6 with 32 bytes of data:
Request timed out.
Reply from 192.168.160.6: bytes=32 time<1ms TTL=127
Reply from 192.168.160.6: bytes=32 time<1ms TTL=127
Reply from 192.168.160.6: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.160.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

T03 - Test from City of London to Factory



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...: 0000.0C3D.3D10
    Physical Address...: FE80::200:CFF:FE3D:3D10
    Link-local IPv6 Address...: ::
    IPv6 Address...: 192.168.231.7
    Subnet Mask...: 255.255.255.240
    Default Gateway...: ::
    DHCP Servers...: 192.168.231.1
    DHCPv6 IAID...: 192.168.249.10
    DHCPv6 Client DUID...: 00-01-00-01-54-55-A5-EE-00-00-0C-3D-3D-10
    DNS Servers...: 192.168.249.3

Bluetooth Connection:

    Connection-specific DNS Suffix...: 00E0.B017.3539
    Physical Address...: ::
    Link-local IPv6 Address...: ::
    IPv6 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: 0.0.0.0
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...: 0.0.0.0
    DHCPv6 Client DUID...: 00-01-00-01-54-55-A5-EE-00-00-0C-3D-3D-10
    DNS Servers...: 192.168.249.3

C:\>ping 192.168.60.5

Pinging 192.168.60.5 with 32 bytes of data:

Reply from 192.168.60.5: bytes=32 time<1ms TTL=126
Reply from 192.168.60.5: bytes=32 time<1ms TTL=126
Reply from 192.168.60.5: bytes=32 time<1ms TTL=126
Reply from 192.168.60.5: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.60.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

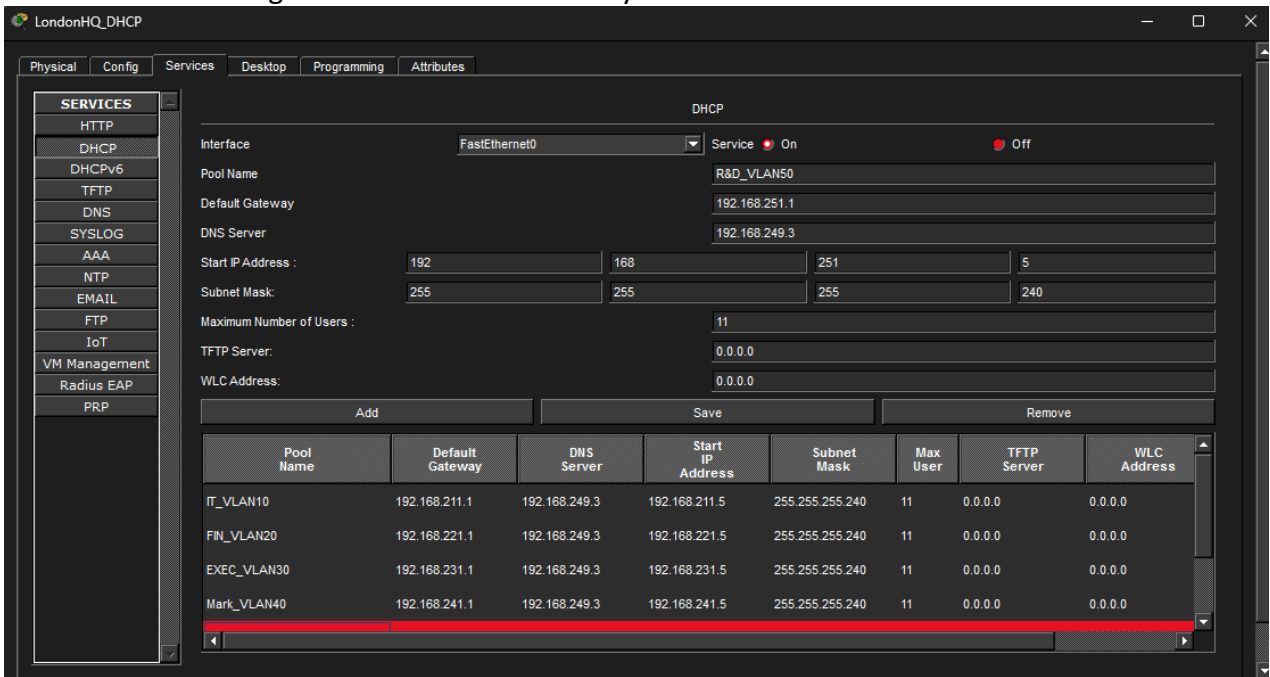
C:\>
```

DHCP Server Test

The objective of these tests is to show DHCP services are functional

T04 - Test from City of London

DHCP server should give out IP address correctly for each VLAN



LondonHQ_DHCP

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: On

Pool Name: R&D_VLAN50

Default Gateway: 192.168.251.1

DNS Server: 192.168.249.3

Start IP Address: 192 168 251 5

Subnet Mask: 255 255 255 240

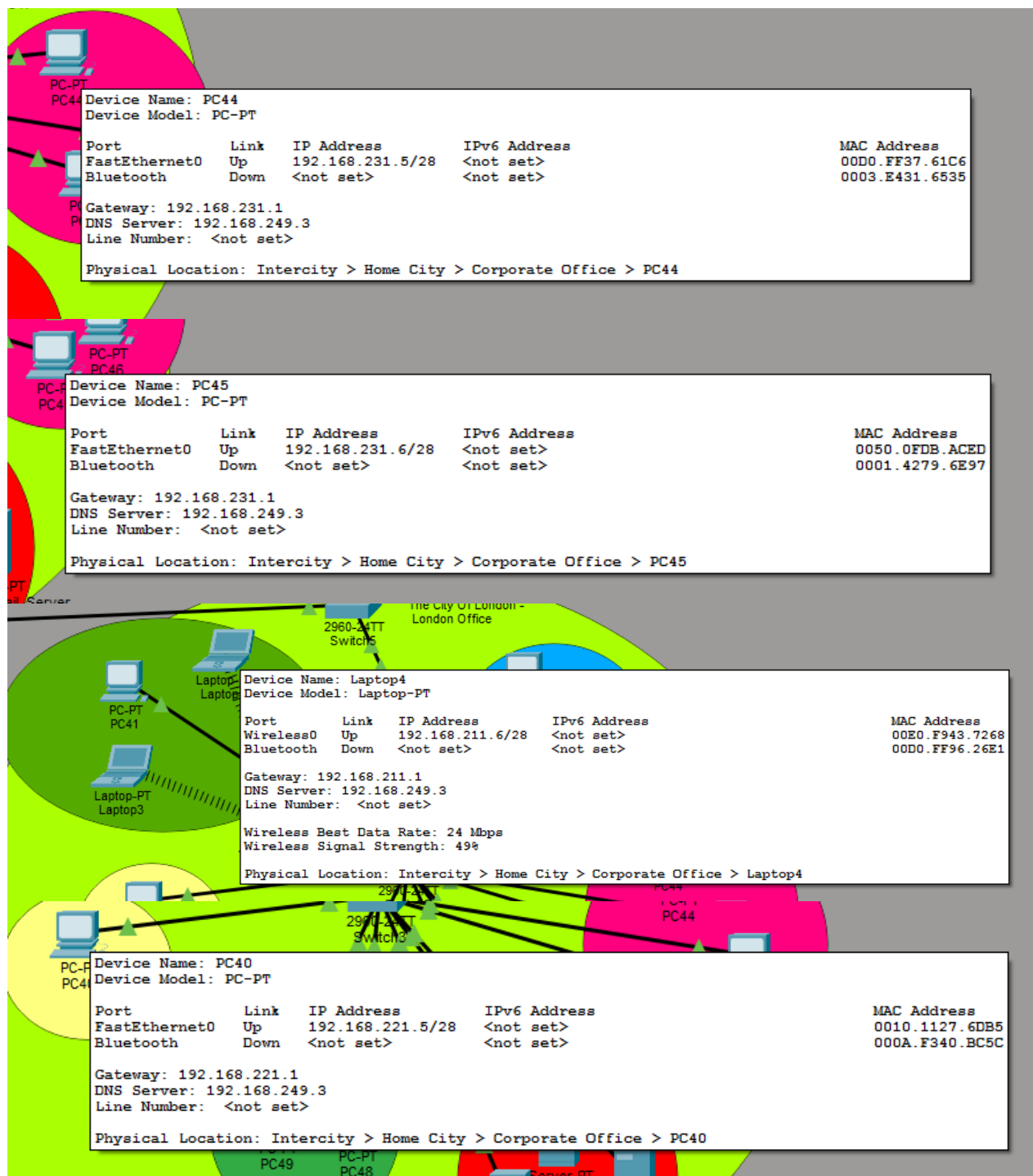
Maximum Number of Users: 11

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
IT_VLAN10	192.168.211.1	192.168.249.3	192.168.211.5	255.255.255.240	11	0.0.0.0	0.0.0.0
FIN_VLAN20	192.168.221.1	192.168.249.3	192.168.221.5	255.255.255.240	11	0.0.0.0	0.0.0.0
EXEC_VLAN30	192.168.231.1	192.168.249.3	192.168.231.5	255.255.255.240	11	0.0.0.0	0.0.0.0
Mark_VLAN40	192.168.241.1	192.168.249.3	192.168.241.5	255.255.255.240	11	0.0.0.0	0.0.0.0



T05 - Test from Factory DHCP server

Factory_DHCP_SERVER

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameserverPool

Default Gateway0.0.0.0

DNS Server0.0.0.0

Start IP Address :192168800

Subnet Mask:255255255240

Maximum Number of Users :16

TFTP Server:0.0.0.0

WLC Address:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
FACTORY_VLAN70	192.168.70.1	192.168.249.3	192.168.70.5	255.255.255.240	11	0.0.0.0	0.0.0.0
ADMIN_VLAN60	192.168.60.1	192.168.249.3	192.168.60.5	255.255.255.240	11	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.80.0	255.255.255.240	16	0.0.0.0	0.0.0.0

PRPAdd

Device Name: PC20

Device Model: PC-PT

Port	Link	IP Address	IPv6 Address	MAC Address
FastEthernet0	Up	192.168.60.5/28	<not set>	0001.C901.0CD7
Bluetooth	Down	<not set>	<not set>	0030.F252.DE5E

Gateway: 192.168.60.1

DNS Server: 192.168.249.3

Line Number: <not set>

Physical Location: Intercity > Home City > Corporate Office > PC20

Wireless connectivity test

T06 - This Test demonstrates wireless connectivity

Laptop9
_ □ ×

Physical
Config
Desktop
Programming
Attributes

Command Prompt X

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

Wireless0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address.....: 0001.63C0.09EC
Link-local IPv6 Address.....: FE80::201:63FF:FEC0:9EC
IPv6 Address.....:
IPv4 Address.....: 192.168.161.7
Subnet Mask.....: 255.255.255.240
Default Gateway.....:
DHCP Servers.....: 192.168.161.1
DHCPv6 IAID.....: 192.168.209.10
DHCPv6 IAID.....: 1777792976
DHCPv6 Client DUID.....: 00-01-00-01-74-88-29-40-00-01-63-C0-09-EC
DNS Servers.....:
DNS Servers.....: 192.168.249.3

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 000C.CF61.33A8
Link-local IPv6 Address.....:
IPv6 Address.....:
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....:
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....: 0.0.0.0
DHCPv6 IAID.....: 1777792976
DHCPv6 Client DUID.....: 00-01-00-01-74-88-29-40-00-01-63-C0-09-EC
DNS Servers.....:
DNS Servers.....: 192.168.249.3

C:\>

```

Server-PT

CANARY_WHAIR DHCP

PC-PT PC33

PC-PT PC34

PC-PT PC35

Laptop9

AccessPoint-PT Access Point2

Port	Link	IP Address	IPv6 Address	MAC Address
Wireless0	Up	192.168.161.7/28	<not set>	0001.63C0.09EC
Bluetooth	Down	<not set>	<not set>	000C.CF61.33A8

Gateway: 192.168.161.1

DNS Server: 192.168.249.3

Line Number: <not set>

Wireless Best Data Rate: 24 Mbps

Wireless Signal Strength: 46%

Physical Location: Intercity > Home City > Corporate Office > Laptop9

```
Laptop2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

Wireless0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address.....: 0060.2F7C.CD7D
Link-local IPv6 Address.....: FE80::260:2FFF:FE7C:CD7D
IPv6 Address.....:
IPv4 Address.....: 192.168.30.7
Subnet Mask.....: 255.255.255.240
Default Gateway.....:
DHCP Servers.....: 192.168.30.1
DHCPv6 IAID.....: 192.168.89.10
DHCPv6 Client DUID.....: 546718090
DNS Servers.....: 00-01-00-01-3E-97-00-E5-00-60-2F-7C-CD-7D
DNS Servers.....: 192.168.249.3

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 0060.47AB.D824
Link-local IPv6 Address.....:
IPv6 Address.....:
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....:
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....: 0.0.0.0
DHCPv6 Client DUID.....: 546718090
DNS Servers.....: 00-01-00-01-3E-97-00-E5-00-60-2F-7C-CD-7D
DNS Servers.....: 192.168.249.3

C:\>ping 192.168.161.7

Pinging 192.168.161.7 with 32 bytes of data:

Request timed out.
Reply from 192.168.161.7: bytes=32 time=22ms TTL=126
Reply from 192.168.161.7: bytes=32 time=37ms TTL=126
Reply from 192.168.161.7: bytes=32 time=37ms TTL=126

Ping statistics for 192.168.161.7:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 37ms, Average = 32ms

C:\>ping 192.168.161.7

Pinging 192.168.161.7 with 32 bytes of data:

Reply from 192.168.161.7: bytes=32 time=9ms TTL=126
Reply from 192.168.161.7: bytes=32 time=15ms TTL=126
Reply from 192.168.161.7: bytes=32 time=11ms TTL=126
Reply from 192.168.161.7: bytes=32 time=23ms TTL=126

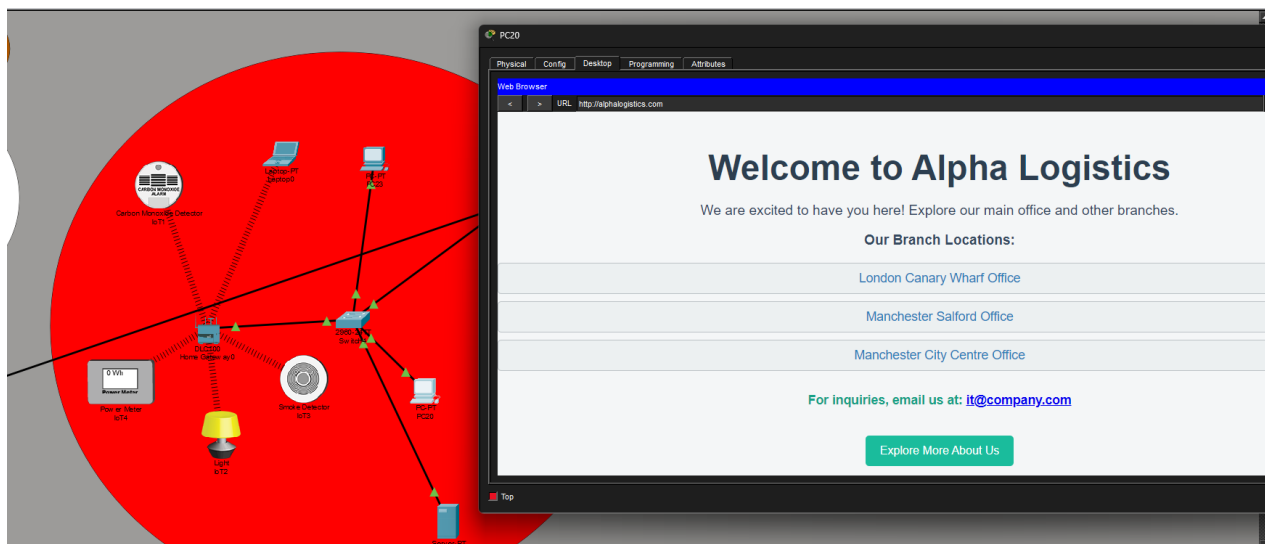
Ping statistics for 192.168.161.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 23ms, Average = 14ms

C:\>
```

WEB and DNS server Test

T07 - Test to see if PC20 can access company website from Factory

As you can see I'm able to access company website from factory using PC20



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping alphalogistics.com

Pinging 192.168.249.3 with 32 bytes of data:

Reply from 192.168.249.3: bytes=32 time=13ms TTL=126
Reply from 192.168.249.3: bytes=32 time=10ms TTL=126
Reply from 192.168.249.3: bytes=32 time=13ms TTL=126
Reply from 192.168.249.3: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.249.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 13ms, Average = 10ms

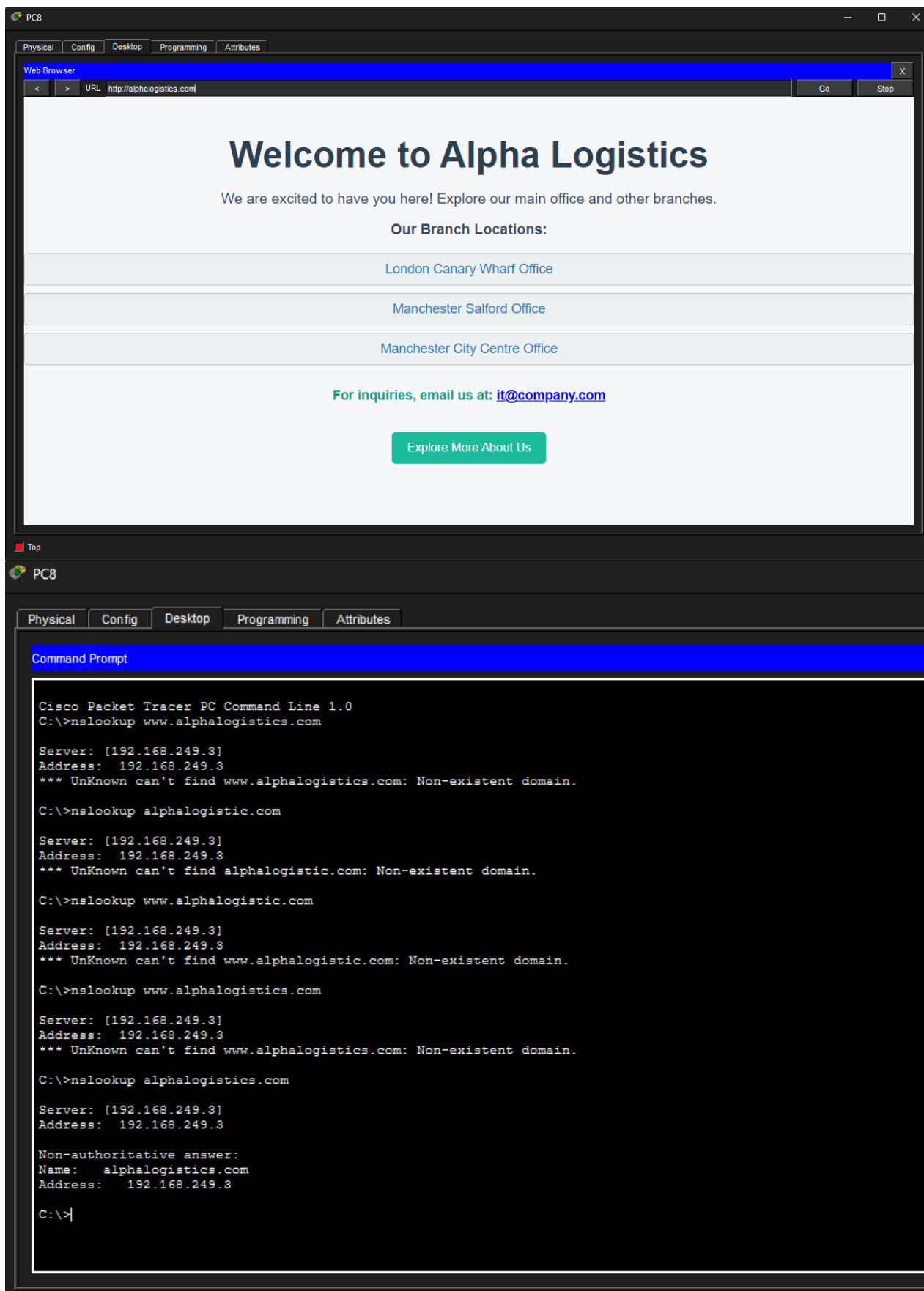
C:\>nslookup alphalogistics.com

Server: [192.168.249.3]
Address: 192.168.249.3

Non-authoritative answer:
Name:   alphalogistics.com
Address: 192.168.249.3

C:\>
```

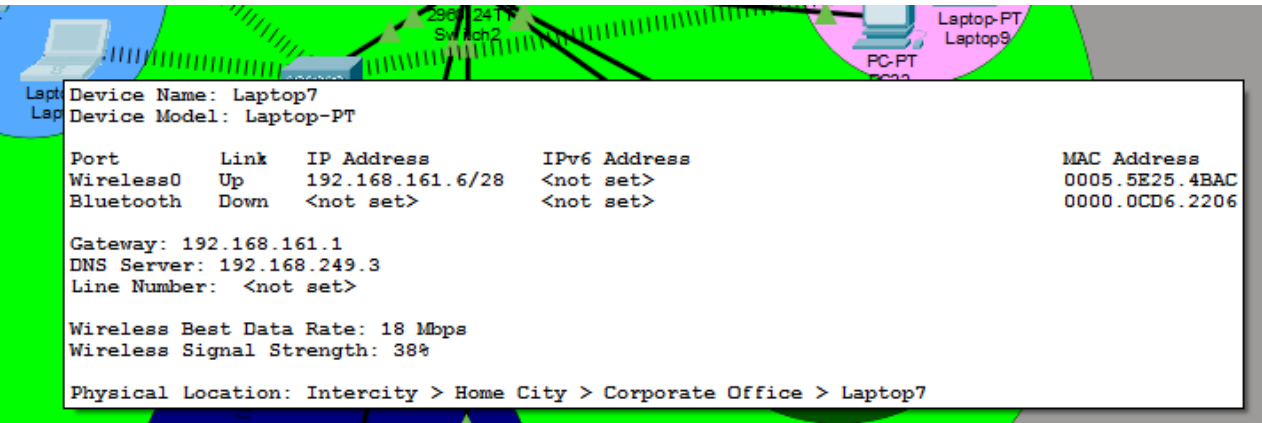
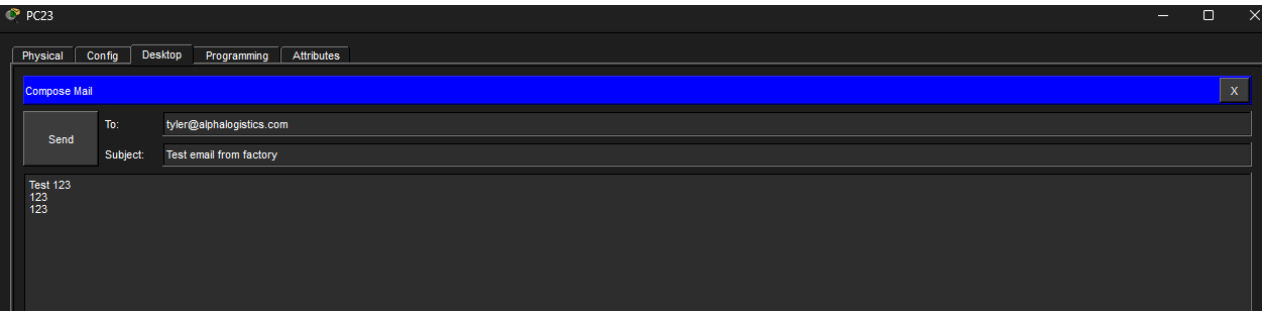
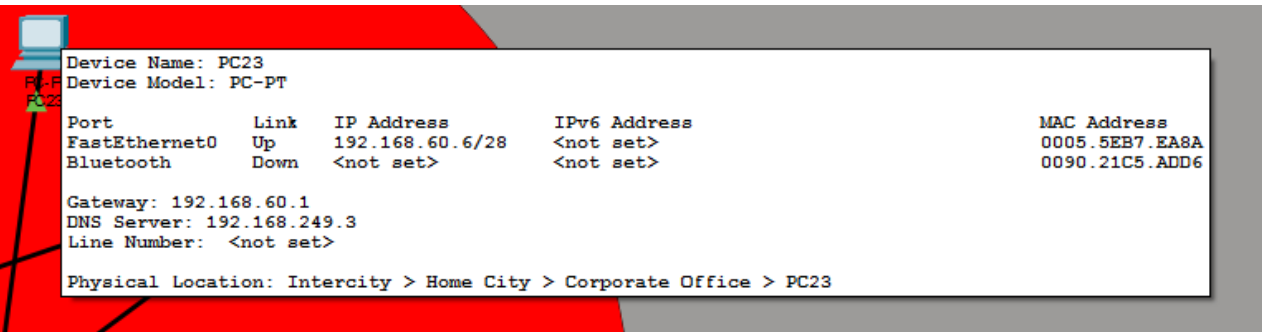
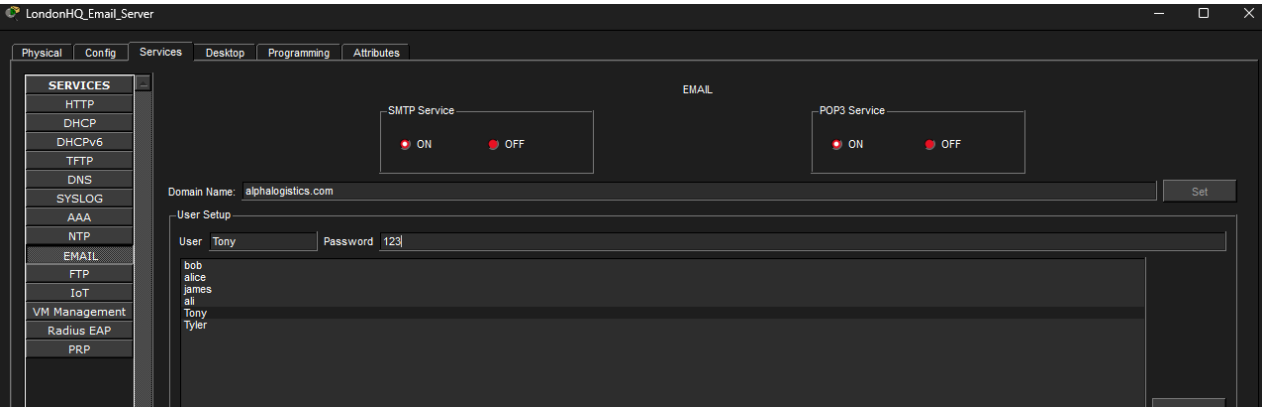
T08 - Test to see if PC8 can access company website from Manchester City Centre
 As you can see I'm able to access company website from Manchester office using PC8

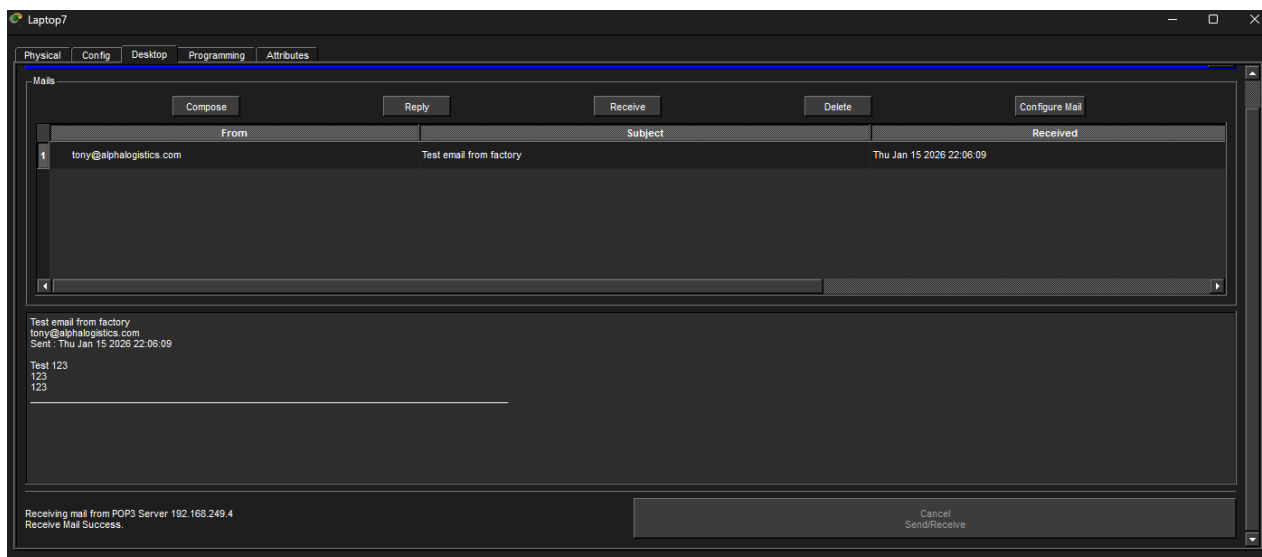


Email Service Test

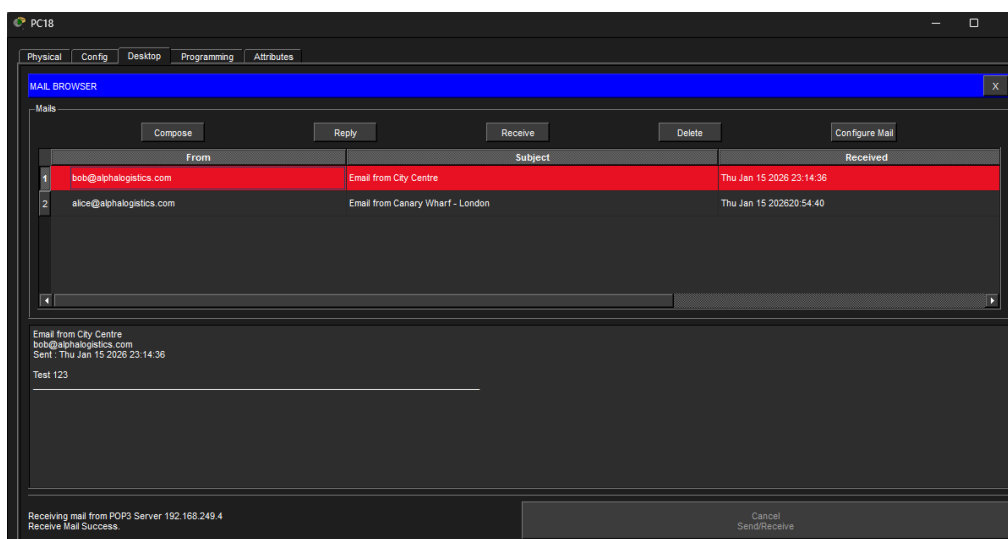
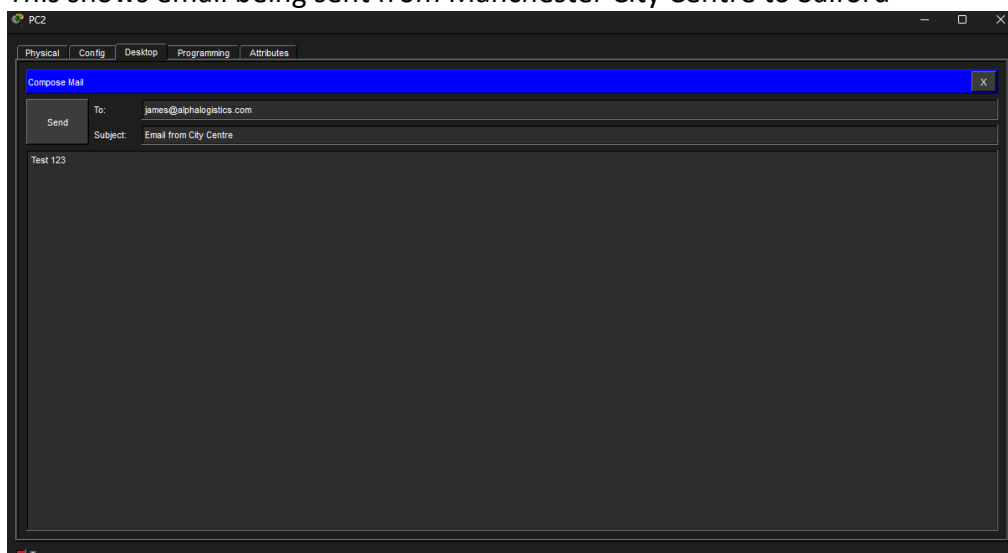
T09 - Test to see if email service works from factory to London

This shows email being sent from factory pc to laptop in London office





T10 - Test to see if email service works from Manchester City Centre to Salford
 This shows email being sent from Manchester City Centre to Salford



Router Configuration Test

T11 shows router from Salford and Factory pinging each other

```
SalfordOffice_Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address          Interface      Hold Uptime    SRTT    RTO    Q    Seq
   (sec)              (ms)          Cnt  Num
0   10.0.0.5          Gig0/0/1      11   00:21:06    40     1000    0    46
1   10.0.0.2          Gig0/0/1      11   00:21:06    40     1000    0    48
2   10.0.0.4          Gig0/0/1      14   00:21:06    40     1000    0    54
3   10.0.0.1          Gig0/0/1      12   00:20:33    40     1000    0    48
```

```
SalfordOffice_Router#ping 10.0.0.4
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.0.0.4, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

```
SalfordOffice_Router#
```

```
FACTORY_ROUTER#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address          Interface      Hold Uptime    SRTT    RTO    Q    Seq
   (sec)              (ms)          Cnt  Num
0   10.0.0.5          Gig0/0/1      13   00:22:23    40     1000    0    46
1   10.0.0.2          Gig0/0/1      12   00:22:23    40     1000    0    48
2   10.0.0.3          Gig0/0/1      12   00:22:23    40     1000    0    48
3   10.0.0.1          Gig0/0/1      12   00:21:51    40     1000    0    48
```

```
FACTORY_ROUTER#ping 10.0.0.3
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.0.0.3, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

```
FACTORY_ROUTER#
```

T12 shows router from City of London and Manchester City pinging each other

```
User Access Verification

Password:
Password:
Password:

LondonHQ>enable
Password:
LondonHQ#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address          Interface      Hold Uptime    SRTT  RTO   Q   Seq
   (sec)              (ms)          (ms)          Cnt  Num
0   10.0.0.2           Gig0/0/1      13   00:24:52    40   1000   0   48
1   10.0.0.3           Gig0/0/1      11   00:24:52    40   1000   0   48
2   10.0.0.4           Gig0/0/1      10   00:24:52    40   1000   0   54
3   10.0.0.5           Gig0/0/1      12   00:24:52    40   1000   0   46

LondonHQ#ping 10.0.0.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.5, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

LondonHQ#
LondonHQ#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

LondonHQ#
```

```
User Access Verification

Username:
Username: admin
Password:

ManchesterCC_Office_Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address          Interface      Hold Uptime    SRTT  RTO   Q   Seq
   (sec)              (ms)          (ms)          Cnt  Num
0   10.0.0.5           Gig0/0/1      11   00:26:12    40   1000   0   46
1   10.0.0.3           Gig0/0/1      14   00:26:12    40   1000   0   48
2   10.0.0.4           Gig0/0/1      13   00:26:12    40   1000   0   54
3   10.0.0.1           Gig0/0/1      14   00:25:40    40   1000   0   48

ManchesterCC_Office_Router#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

ManchesterCC_Office_Router#
```

Network Evaluation and Discussion

Reliability and Redundancy

- The network uses a central branch router in city of London that interconnects all offices via a routed core segment using EIGRP. The design ensures that there is consistent connectivity between all offices and enables dynamic route recalculation if a router becomes unavailable. However, the current network design contains several weaknesses such as that main branch router and core switch is a critical component if they were to fail it would lead to disrupt inter-site communications also each office location has one WAN connection to main branch which means if main router was to fail it would cause isolation to all offices

Scalability

- The network has been designed with VLAN segmentation and CIDR-based subnetting, which supports logical growth. Each department is assigned its own VLAN and subnet, reducing broadcast traffic and simplifying management. The addressing scheme allows additional subnets to be introduced easily, and EIGRP provides scalable routing suitable for larger network. However, some VLANs use /28 subnet (255.255.255.240) which limits the growth to 14 hosts per department, a bigger size subnet is needed such as /27 or /26 if the staff increases significantly

Performance

- Performance has been enhanced through VLAN segmentation which isolates broadcast domains and improves network efficiency. Inter-VLAN routing is handled by router sub-interfaces using 802.1Q trunking allowing more secure communication. Centralised services such as DNS, web hosting and email are in the server VLAN which simplifies administration. Wireless connectivity is implemented through access points mapped to a dedicated staff VLAN enabling controlled mobility without exposing internal networks.

Cost, Future and Security Considerations

- The design priorities cost efficiency while maintaining acceptable operational reliability full redundancy at every layer would require additional routers, switches and WAN links which would increase expenses significantly, given that the client size, the current solution offers a practical compromise between cost and availability
- Security has been incorporated by enabling passwords, VLAN segmentations, and 1 privileged admin. However, in future the security can be improved further by implementing ACLs, port security and implementing dedicated firewalls to prevent harmful traffic entering the networks

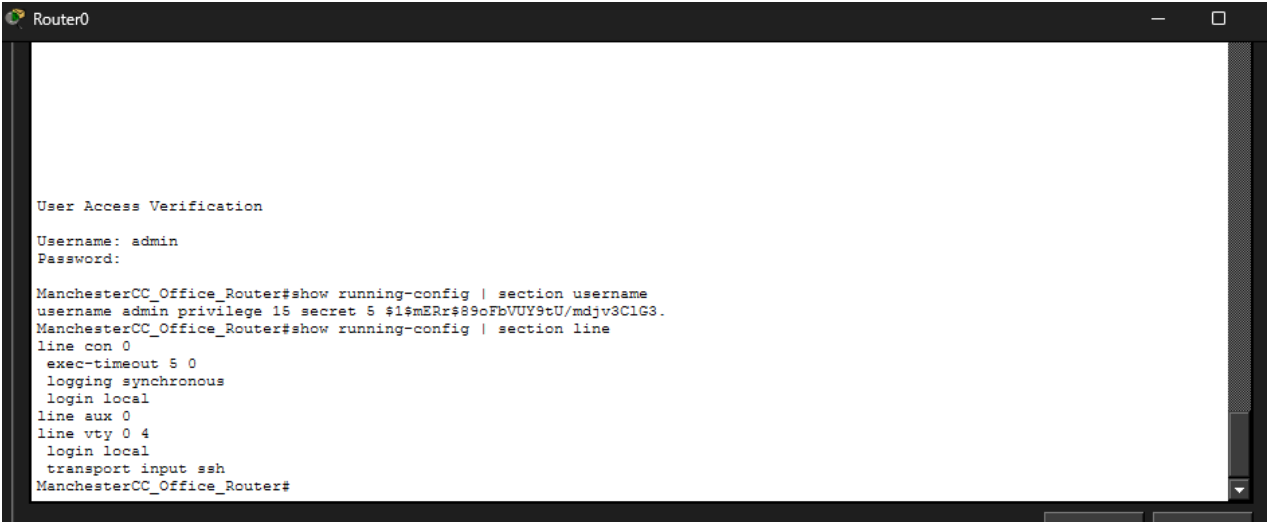
Conclusion

Overall, the network design successfully meets the functional requirements of ALPHA-Logistics by providing reliable inter-office connectivity, structured addressing, dynamic routing and centralised services. While the solution is not fully redundant, it is scalable, cost-effective and suitable for current operational needs, with clear upgrade paths available for future growth and security

Appendix A

Routers security configuration

Manchester – City Centre

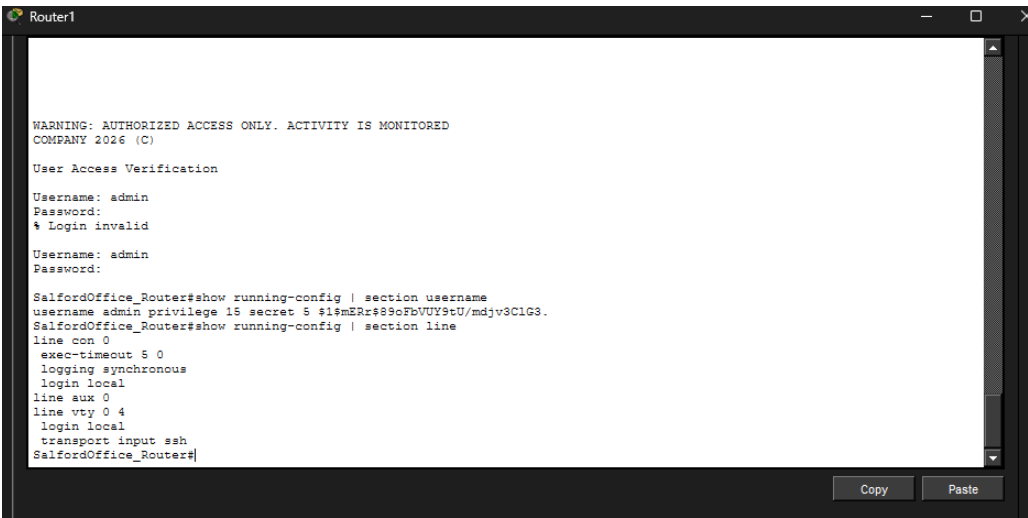


```
Router0

User Access Verification
Username: admin
Password:

ManchesterCC_Office_Router#show running-config | section username
username admin privilege 15 secret 5 $1$mERr$889oFbVUY9tU/mdjv3ClG3.
ManchesterCC_Office_Router#show running-config | section line
line con 0
exec-timeout 5 0
logging synchronous
login local
line aux 0
line vty 0 4
login local
transport input ssh
ManchesterCC_Office_Router#
```

Manchester – Salford



```
Router1

WARNING: AUTHORIZED ACCESS ONLY. ACTIVITY IS MONITORED
COMPANY 2026 (C)

User Access Verification
Username: admin
Password:
% Login invalid

Username: admin
Password:

SalfordOffice_Router#show running-config | section username
username admin privilege 15 secret 5 $1$mERr$889oFbVUY9tU/mdjv3ClG3.
SalfordOffice_Router#show running-config | section line
line con 0
exec-timeout 5 0
logging synchronous
login local
line aux 0
line vty 0 4
login local
transport input ssh
SalfordOffice_Router#
```

London – Canary Wharf

```
Router2

User Access Verification

Username: admin
Password:

CanaryWharf_Router#show running-config | section line
line con 0
  exec-timeout 5 0
  login local
line aux 0
line vty 0 4
  login local
  transport input ssh
CanaryWharf_Router#show running-config | section username
username admin privilege 15 secret 5 $1$mERr$89oFbVUY9tU/mdjv3C1G3.
CanaryWharf_Router#
```

London – The City of London office

```
Router4

User Access Verification

Password:
Password:

LondonHQ>enable
Password:
LondonHQ#show running-config | section username
username superadmin privilege 15 secret 5 $1$mERr$je0fk4Wj/0sYY6zb.vf.3.
LondonHQ#show running-config | section line
line con 0
  password 7 08725C67220E56404424
  login
line aux 0
line vty 0 4
  password 7 08177877584B56
  login local
  transport input ssh
LondonHQ#
```

References

No	References
1	<i>Rethinking hub-and-spoke logistics: Can direct-to-consumer be more cost-effective? (2025) Argon & Co. Available at: https://www.argonandco.com/en/news-insights/articles/rethinking-hub-and-spoke-logistics-can-direct-to-consumer-be-more-cost-effective/ (Accessed: 8 January 2026).</i>
2	<i>VPC CIDR Blocks (2025) CoreWeave. Available at: https://docs.coreweave.com/docs/products/networking/vpc/vpc-cidr (Accessed: 12 January 2026).</i>
3	<i>Network segmentation vs. Vlan explained (no date) AlgoSec. Available at: https://www.algosec.com/blog/network-segmentation-vs-vlan (Accessed: 12 January 2026).</i>
4	<i>Introduction to eigrp (2025) Cisco. Available at: http://cisco.com/c/en/us/support/docs/ip/enhanced-interior-gateway-routing-protocol-eigrp/13669-1.html (Accessed: 14 January 2026).</i>
5	<i>(No date) Homeworkforyou. Available at: https://www.homeworkforyou.com/static_media/uploadedfiles/1707230158_1668475__959..pdf (Accessed: 8 December 2025).</i>
6	
7	